Nearly half of junction-spanning gapmer designs against the *NR4A3* fusions of extraskeletal myxoid chondrosarcoma pair a wild-type parent gene, and a longer catalytic gap cannot separate them

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Running title. Junction gapmers across NR4A3 fusions

Keywords. antisense oligonucleotide; gapmer; RNase-H1; fusion transcript; NR4A3; extraskeletal myxoid chondrosarcoma

Abstract

Extraskeletal myxoid chondrosarcoma (EMC) is an ultra-rare sarcoma in which a variable partner gene fuses to *NR4A3*. Its breakpoint junction is present in no normal gene, so an antisense gapmer could in principle cleave the fusion transcript while sparing both parent genes; none has been reported, and designs are easy to find. Of 190 candidates across the 38 in-frame junctions of the five known partners, 87 pair their catalytic gap against a mature parent transcript over a contiguous duplex of at least ten base pairs, 61 against healthy *NR4A3*; another 19 pair it in precursor RNA, where RNase-H1 is also active and a mature-transcript screen cannot reach. Arbitrary sequence does not: scrambles reach 6.2% on the same screen and chimeras joining the same two parents at random offsets 23.8%, against 45.8% observed. Designs that look clean mostly stop looking clean: six of the nine with no sense-strand *near-match* (≥14 of 16 positions) lose the property at ten times the search depth. Lengthening the gap cannot help, for an arithmetic rather than an empirical reason: in every design of all three geometries tiled here, the junction-unique bases a longer gap wins and the wild-type-parent duplex it concedes are the same nucleotides. Two reagents are named for synthesis with their off-target loads, 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6, together reaching 68.4% of molecularly confirmed cases (39.9–82.8%); three as not to be used because each pairs its whole catalytic gap against the patient's own un-rearranged *NR4A3* allele, with the controls and the selectivity value that would falsify the ranking. The design and screening pipeline is released, so a reagent can be designed for a breakpoint outside this panel by the same procedure. The work is computational: no wet-lab experiment was performed, no sequence named has been synthesised or tested, and nothing here asserts efficacy, safety, delivery to a tumour or clinical readiness.

1 · Introduction

EMC is defined in the large majority of cases by an in-frame fusion of *EWSR1* to the orphan nuclear receptor *NR4A3*,[1] with *TAF15* a substantial minority and *TCF12* and *TFG* rare.[2] *FUS::NR4A3* is reported in a recent series that identified it by sequencing in two of five variant EMCs.[3] Next-generation sequencing of six EMCs finds few recurrent secondary mutations beyond the fusion,[4] so it is to a first approximation the single clonal driver. In every junction type described, the predicted product joins the partner's amino-terminal transactivation domain to essentially the entire NR4A3 protein, including its nuclear-receptor DNA-binding domain.[1,5]

That driver is currently untargeted. Surgery with clear margins is the backbone of localised disease, and for advanced disease no clinically validated agent directly targets *NR4A3*.[6] The largest EMC-specific prospective study, a single-arm phase 2 of pazopanib in centrally confirmed *NR4A3*-translocated disease, returned four objective responses in 22 evaluable patients;[7] anthracycline-based chemotherapy returned four in ten evaluable patients in a molecularly confirmed retrospective series of eleven, a result that series presents as running counter to the prior record.[8] Two of those sources report a low objective response rate to that chemotherapy and low sensitivity to cytotoxic chemotherapy generally,[6,7] and none of the three publishes a response rate by line of therapy. The population a fusion-directed agent would address is close to the whole disease: across 58 molecularly confirmed cases, 79% carried *EWSR1::NR4A3*, 16% *TAF15::NR4A3* and 3% *TCF12::NR4A3*.[9]

The chimeric mRNA offers a discrimination handle its protein product does not. The *NR4A3* ligand-binding domain is retained near-intact and identical in sequence to wild-type NR4A3, so a ligand that engages it cannot distinguish fusion from wild type. The breakpoint junction can: it is a contiguous stretch of sequence present in no normal transcript and absent from both parent transcripts, so selectivity can in principle be enforced by base-pairing rather than by protein conformation. It is the target of every design here.

Targeting a fusion breakpoint with an oligonucleotide is not new. The approach has a continuous lineage from 1991,[10] including RNase-H-dependent antisense at a sarcoma fusion breakpoint in 1997,[11] and the fusion-exclusivity rationale was stated as a general principle in 2005.[12] Parental sparing has been reported at four fusions, two with a readout on the wild-type parent transcript and two asserting junction specificity without one.[13–16] A bi-shRNA lipoplex against the *EWSR1::FLI1* junction reached preclinical justification,[17] and a GalNAc-conjugated junction siRNA in fibrolamellar hepatocellular carcinoma passed the delivery gate in a rare fusion-driven cancer.[18] The contribution here is the indication rather than the modality: across 5,153 unique records retrieved from Europe PMC, four mention *EWSR1::NR4A3* at title or abstract level, resolving to three papers, none an oligonucleotide study.

Two questions follow that the field has not asked of this disease. Prior design work has addressed only *EWSR1*, while the partner varies, and partner identity may not be clinically inert: across the two series of antiangiogenic tyrosine-kinase inhibition in advanced EMC that report a partner breakdown at all, no objective response is reported in a *TAF15* patient, on a *TAF15* arm of three to five patients whose Wilson upper bound remains compatible with equal response.[7,19] Neither primary report states the per-arm denominators; the sunitinib report states the partner split only qualitatively, and both denominators are read from published reviews of the two series.[20,21] The second is whether a junction oligonucleotide must be bespoke per patient, or whether one sequence can serve more than one fusion, which decides whether the deployable artefact for an ultra-rare disease is a stock reagent or a panel.

This paper answers both from sequence, sets out where a computational screen stops being able to answer them, and ends by naming the oligonucleotides to synthesise, the controls that make a knockdown result interpretable, and the pre-registrable threshold that would falsify the ranking every candidate here is ordered by (§4). The design and screening procedure is released with the artefacts, so a candidate can be designed by the same route for a breakpoint the panel does not carry (§4.5). Box 1 collects the sequences, the cautions and the terms of art the sections below use before the Methods (§6) define them.

Box 1 · Sequences, cautions and the void condition

Every line below points at a fuller statement in the section cited, and none of it is argued here.

The terms of art (§6). Each design is one *register* of its junction, one way of sliding the 16-mer while the breakpoint still falls inside the six-nucleotide DNA *catalytic gap* RNase-H1 cleaves within; the 5-6-5 geometry admits five per junction. A design's *gap-level margin* is the count of junction-unique bases inside that gap on the shorter side of the breakpoint, and every ranking here is built on it. A *near-match* is a transcript window pairing a design at 14 or more of its 16 positions, and is *gap-paired* where the six gap positions are themselves paired, which is the class RNase-H1 could cleave. A design is *clean* where a complete hit list at a stated search depth returns no sense-strand near-match. Five screens are applied, numbered here as in §6: (1) the alignment screen, (2) the exhaustive transcript scan, (3) the pre-mRNA screen, (4) the mature-parent screen and (5) the genome scan.

The two reagents to synthesise (§4.1, Table 4). 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 joined to *NR4A3* exon 3, carrying 123 gap-paired sense-strand near-matches at six gene loci at the deeper search ceiling together with a sense-strand near-match in wild-type *TAF15* precursor RNA (§4.3); and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6, carrying 8 such near-matches at five loci and no sense-strand pre-mRNA site. Both hold the top gap-level margin of 3, and neither has been synthesised or tested. Between them they reach roughly two thirds of molecularly confirmed cases (Table 7).

Three designs not to be used (§2.6). 5′-CAGTGGGCTCTCCACG-3′ and 5′-GCAGTGGGCTCTCCAC-3′ at *EWSR1* exon 13 joined to *NR4A3* exon 2, and 5′-TGATGAGGGCCTTGTG-3′ at *TAF15* exon 6 joined to the *NR4A3* intron-2 cryptic exon. Each cleared the spliced-cDNA parent screen, and each pairs its whole catalytic gap against the patient's own un-rearranged *NR4A3* allele, which that screen cannot see.

The cell line (§3). No *NR4A3* fusion is detectable in H-EMC-SS on the public record, so no reagent named here can be tested in it. That is not a statement that the line is misidentified.

The replicate floor and the void condition (§4.4). Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose–response in the same wells, and the cut is 5.0. Three biological replicates are a floor and not a target: above a replicate standard deviation of about 0.65 on the log scale, no observed ratio can place a 95% upper bound below that cut at three replicates, so the test cannot fail and the design is void rather than negative. The controls, the assay placement and the limit-of-quantification condition without which the ratio is not reportable are in §4.4.

A design procedure for a breakpoint outside this panel (§4.5). The pipeline that produced the 190 designs is released with the artefacts, and §4.5 gives its input, the five screens a new design must clear, and the limit on what it produces: a candidate, not a validated reagent.

2 · Results

The results are ordered for a laboratory deciding what to make. How far any of the counts below can be trusted is stated in §5, which bounds all of them.

How the counts are denominated. Six numbers recur below and they are not interchangeable. 231 is the donor-exon by acceptor-exon pairs graded for frame. 38 are the in-frame junctions among them. Those 38 carry 190 design records, which are 176 distinct molecules, because nine of the 16-mers span more than one partner's junction and are recorded once per junction. Of the 190, 183 have a returned specificity screen; the other seven failed at the remote service, which matters because a free energy needs only a sequence where a screen needs a query that came back. 187 is the count re-screened at the tenfold deeper ceiling. Each result below names the denominator it uses.

2.1 · The reading frame as the bound on junction space

Grading all 231 donor-exon by acceptor-exon pairs across the five partners returns 38 in-frame junctions (Table 1, Figure 1). The refusals are structural. *NR4A3* exon 2 carries no coding sequence and is refused in every pair, and an exon-4 acceptor would delete the *NR4A3* DNA-binding domain that every reported EMC chimera retains. All the variance therefore sits in the exon-3 column. Within that column, being in frame reduces to a single arithmetic condition, a donor coding phase of 1, which is necessary and sufficient across its 77 rows but only necessary across all 231.

Among these, *EWSR1* exon 12 joined to *NR4A3* exon 3 is the junction reported most often: type 1 in 10 of the 15 *EWSR1*-rearranged tumours of an 18-case series.[22] Designs at this junction therefore correspond to the largest documented patient group.

No design at any of the 38 junctions is a perfect complement of any of the six parent transcripts. That test excluded none of the 190, because a junction-spanning window cannot occur intact in a parent. GC runs 25.0–75.0% across partners, with 132 of the 190 inside the conventional 40–60% band. Finding candidate sequences is therefore not the constraint on this modality in this disease.

2.2 · Cross-partner coverage by a single oligonucleotide

Nine designs span more than one partner's junction exactly, and all nine draw from *EWSR1*, *TAF15* and *FUS* (Figure 3). Five cover the same three-partner set, differing only in register across the junction. The best by gap-level margin is 5′-GGGCATATCATCAAAC-3′ (43.8% GC, gap-level margin 3), which divides eight donor and eight acceptor bases at the junction of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, and occurs in none of the six wild-type parent transcripts. The basis is sequence identity: the three donors are identical over the ten bases immediately 5′ of their breakpoints, diverging at the eleventh. No design draws more than ten donor bases, which is what makes the coverage arithmetically possible.

In one respect the published data contradict the clinical reading of this result. The only exon-resolved *TAF15::NR4A3* breakpoints reported in EMC are at exon 6, not exon 11. The primary report of the variant fusion places the breakpoint there,[23] and in a series of 18 EMCs all three *TAF15*-rearranged tumours carried exon 6 joined to *NR4A3* exon 3.[22] The exon-6 junction shares a single donor base with the exon-11 junction, so this oligonucleotide cannot engage the *TAF15* junction that patients are reported to carry.

That junction is itself in-frame and yields five fusion-specific designs (43.8–50.0% GC), all five screened and orientation-filtered. Every one of them retains a sense-strand near-match spanning the catalytic gap. At the tenfold deeper ceiling, where every hit list is complete, those recount to three gene loci at best, and five for the design its gap-level margin ranks first, three of those five annotated only as predicted gene models (Table 4). Two of the five nonetheless return no exact and no single-mismatch match on the exhaustive transcript scan.

So the one *TAF15* junction with a published breakpoint is designable and is not among the cleaner junctions, while the junction the multi-partner result rests on has no reported patient. For *FUS* no exon-resolved EMC breakpoint has been published at all. The three-partner result is therefore a statement about FET-family sequence architecture and a hypothesis about junctions not yet observed. It is not a claim that one reagent serves three patient groups. Testing it requires breakpoint sequencing of archival *TAF15*- and *FUS*-positive cases.

2.3 · The non-FET partners: coverage and specificity

TCF12 and *TFG* are the partners in this panel that are not FET-family proteins, and neither appears in any of the nine multi-partner sets: all nine draw only on *EWSR1*, *TAF15* and *FUS*. *TCF12* reaches multi-partner coverage only under a relaxed criterion that tolerates mismatches in the oligonucleotide wings. That check had little power to fail, since any non-homologous donor would be excluded, so it does not separate FET paralogy from incidental exon homology. The stronger evidence for paralogy is that four additional two-partner sets are also FET-only.

Specificity does not sort by partner. Taking at each junction the lowest count any of its designs achieves after the orientation filter — a per-design minimum, which is in the released per-junction screens and not in Table 2, whose row is that junction's highest-margin design and not its cleanest — every one of the five partners has at least one junction whose best design carries no sense-strand near-match across the catalytic gap at the default search ceiling: three of eight at both *TCF12* and *FUS*, two of eight at *EWSR1*, one of eight at *TAF15* and one of six at *TFG*. At the tenfold deeper ceiling that becomes four partners: each of *EWSR1*, *FUS*, *TAF15* and *TCF12* keeps one such junction and *TFG* keeps none, its single default-depth zero returning 29 across the gap when searched deeper. The minima therefore separate junctions rather than partners. Which exon a fusion breaks at matters more for specificity than which gene it breaks into.

The same tension the *TAF15* result carries applies to *TCF12*, and in the same direction. The one published *TCF12::NR4A3* breakpoint describes a chimera retaining the first 108 TCF12 residues,[5] and names no exon; the same authors deposited the chimeric cDNA, and that deposit resolves the junction to the nucleotide. GenBank AF289510.1 carries two chromosome-tagged source features that meet at the junction. Mapped against the transcript models used here, the donor side ends at *TCF12* exon 5 and at no other exon, the acceptor side begins at *NR4A3* exon 3 and at no other exon, and the twelve bases either side of the seam are identical to the seam the panel was designed on. The base-level assignment, its chromosome-tagged coordinates and the translation check that reproduces the deposit's own recorded protein are in the released artefact. That junction is in-frame and designable, and its best-margin design retains 17 gap-spanning near-matches at the deeper ceiling, every one of them a variant of a single curated locus, *PIK3CG* (Table 4). None of the four *TCF12* designs with no sense-strand near-match is at that exon. So for *TCF12* as for *TAF15*, the junction a patient is reported to carry is designable and is not among the clean ones, while the clean junctions have no reported patient. What remains unmeasured at *TCF12* is not the exon but the distribution: one *TCF12*-rearranged tumour has been sequenced at this junction, and it is the tumour the junction was defined by. No search of the nucleotide or read archives returns a second *TCF12::NR4A3* sequence, and what that leaves the coverage estimate resting on is stated with the estimate (§4.1).

The same archive search resolves *TFG*, the other non-FET partner, in the same way and to the same limit. No paper places a *TFG::NR4A3* breakpoint at an exon, but a deposited chimeric mRNA record does — GenBank AY532911.1, annotated as a *TFG-NR4A3* fusion protein — and it lands on *TFG* exon 7 joined to *NR4A3* exon 3, a junction this panel already carries. Four patent sequence records agree at the seam, corroborating a sequence rather than four patients, being one family from one group. As at *TCF12*, what the deposit supplies is the exon and not the distribution: no source states what fraction of *TFG*-rearranged tumours break there, and *TFG* does not appear in the partner counts of the 58-case cohort every coverage figure here is denominated on, so this changes which junctions are reported and no percentage.

2.4 · Strand orientation, and designs with no sense-strand near-match

All 38 in-frame junctions were screened with orientation filtered, covering 183 designs, and Table 2 gives the per-junction result. Of the 1,677 apparent cleavage risks across the retained hit lists, 738 sit on the minus strand, or 44%. An antisense oligonucleotide cannot base-pair with those at all.

The proportion is not uniform. It runs from 0% at *TFG* exon 4, where no apparent risk is minus-strand, to 100% at both *EWSR1* exon 1 and *TCF12* exon 7, where every one is. That non-uniformity is what makes the filter worth applying rather than approximating. A uniform inflation would rescale every junction and leave their ordering intact; this one reorders them. *EWSR1* exons 7 and 13 return 55 and 57 apparent gap-spanning hits, and after filtering they stand at 6 and 53.

After filtering, nine designs at six junctions carry no sense-strand near-match among non-parent transcripts (Table 3), spanning four of the five partners: three at *EWSR1* exon 1 (5′-GGGCATATCCGTGGAC-3′, 5′-GGCATATCCGTGGACG-3′, 5′-GCATATCCGTGGACGC-3′), one at *FUS* exon 8 (5′-AGGGCATATCGGAGTC-3′), one at *TAF15* exon 1 (5′-GGGCATATCCGACATG-3′), and four at *TCF12* — 5′-GGGCATATCTCTATAA-3′ at exon 17, 5′-CAGGGCATATCTTGCA-3′ at exon 9, and 5′-GGCATATCAAGCGCTG-3′ and 5′-GCATATCAAGCGCTGC-3′ at exon 7. The exhaustive transcript scan agrees independently: each returns no exact and no single-mismatch match anywhere in 186,185 transcripts. The two screens fail in different ways, so their agreement is not a restatement. One is a heuristic alignment search over both strands; the other an exhaustive substitution scan over the sense orientation only. The pre-mRNA screen, over a compartment neither of those reaches, does not overturn them either: none of the nine has a sense-strand site in parent pre-mRNA (§2.5).

The graded re-score agrees, with one instructive exception. Scoring every retained hit by the residual cleavage a gap-internal mismatch is predicted to permit, under both literature bounds, returns a residual load of zero for all nine. It returns zero for one further design too, at *FUS* exon 11, which is not counted as clean here. That design returns 21 near-matches, of which only 15 are retained, and all 15 are minus-strand. The graded score therefore sees nothing it can score, while the cleanliness criterion refuses the design because the strand of the six unretained hits is unknown. The graded model has no censoring guard, so it can award a zero the hit list does not support, and the stricter count is the one reported. A zero for the nine is arithmetic rather than an independent measurement: it follows from their having no sense-strand hit to score.

Every junction was then re-screened at a tenfold deeper alignment ceiling, with retention raised to match so that no hit list is truncated: 38 junctions and 187 design records. The result withdraws most of the set above. The 187 are the panel's 190 less three that failed at the remote service on this pass, two at *FUS* exon 5 and one at *TFG* exon 2. Each had already returned 23, 41 and 31 near-matches at the default depth, so none was a candidate and no count below depends on them. Only three of the nine still carry no sense-strand near-match: 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8, 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 and 5′-GGCATATCAAGCGCTG-3′ at *TCF12* exon 7, each of which returned the same count at both depths. The other six did not. The three *EWSR1* exon-1 designs had returned no near-match at all at the default ceiling and return 27, 29 and 84; 5′-GGGCATATCTCTATAA-3′ at *TCF12* exon 17 goes from 8 to 118, and 5′-CAGGGCATATCTTGCA-3′ at *TCF12* exon 9 from 7 to 67. Three of the six carry hits that span the catalytic gap and so are cleavage risks rather than merely sense-strand matches: 64 for 5′-GCATATCCGTGGACGC-3′, 14 for 5′-GGGCATATCTCTATAA-3′ and 11 for 5′-CAGGGCATATCTTGCA-3′. A count of zero at the default ceiling was not a count of zero, which is the sharpest form of the bound §5 sets out.

The deeper pass also decided what the default one could not. Seven of the 190 designs had failed at the remote service and carried no count at all — a different seven from the seven §5 reports as withheld by retention alone, which do carry default-depth counts; all seven returned at the deeper ceiling, six of them dirty and one — 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7 — with three near-matches and none on the sense strand. So the set of designs with a complete hit list and no sense-strand near-match is four at this depth rather than three: a design the shallower pass never screened joins the three that survived it. The deeper counts are reported as their own measurement and no figure quoted above is restated from them.

The orientation call is corroborated independently of any of this. Ten designs return perfect 16/16 BLAST matches while the sense-only exhaustive scan reports no exact match. Both results can only be correct if every one of those BLAST hits is on the minus strand, and every one is.

2.5 · The parents: liability in pre-mRNA and in mature transcript

RNase-H1 is active in the nucleus and gapmers engage pre-mRNA, so a screen over mature transcripts cannot see intronic or intron–exon-spanning sites. That omission is not neutral in its direction. A junction gapmer's two halves are both exonic, and in a parent pre-mRNA an exon is followed by an intron rather than by the next exon. Parent pre-mRNA is therefore precisely where a design's donor half sits beside sequence no mature screen has compared it against. A mature-only screen returns a low count partly by construction.

Of the 190 designs, 53 have a near-match somewhere in parent pre-mRNA. Nineteen carry one that meets all three conditions that would make it dangerous: it is on the sense strand, it pairs the catalytic gap in full, and it touches intronic sequence. That third condition is what makes such a site invisible to both transcript screens, rather than a re-count of something already reported.

The step from 53 to 19 is a threshold rather than a measurement, and the class it removes is the one the Methods (§6) decline to dismiss. Forty designs carry a sense-strand parent pre-mRNA site. The 19 counted here are those pairing the catalytic gap in full; the remaining 21 pair all of it but one or two positions. Of their 28 sites, 26 are a single gap mismatch short, and five are in *NR4A3* itself. Under the bounds this work adopts, a single mismatch inside the gap does not abolish cleavage, so those 21 are not a null result. They are excluded because a graded count over this compartment would need a discrimination model the literature does not supply for a parent duplex. The same condition governs the mature-parent screen below, which considers only windows pairing the whole gap. Every count in this section should be read as the fully-paired class, not as the whole parent liability.

Those 19 sites fall into two classes that do not mix, and only one is mechanistically interesting. Nine are intron–exon-spanning, and every one is in *NR4A3* at the same place: six or seven nucleotides into intron 2, spanning the boundary into exon 3. That follows from the design problem. A junction gapmer's acceptor half is the 5′ end of *NR4A3* exon 3, and the wild-type *NR4A3* transcript reaches that same exon across its own splice junction. So a design whose donor half also matches the 3′ end of intron 2, within the mismatch budget, pairs across the real splice site. That is a route to wild-type *NR4A3* engagement which does not pass through the fusion at all, in the compartment where RNase-H1 is active. It is the discrimination question this paper is about. The other ten are wholly intronic and every one is in *TCF12*, which contributes 365,096 of the 517,157 intronic nucleotides searched. That is 71% of the search space accounting for 100% of the class, which is what sequence volume alone predicts and should not be read as anything about *TCF12*.

The liability tracks the tiling register, of which the gap-level margin is a function. At margin 1, 12 of 76 designs carry a pre-mRNA site; at margin 2, 7 of 76; at margin 3, none of 38. Eight of the nine *NR4A3* boundary sites are at the shortest donor-side register, which needs the fewest intronic bases to match. None of the nine designs with no sense-strand near-match on either transcript screen carries one.

The second class is in mature parent transcript, and it is larger. Each of the first three screens misses it for its own reason. The alignment screen excludes parent records by design and filters at ≥14/16 identity. The exhaustive transcript scan admits only one mismatch. The pre-mRNA screen searches unspliced sequence and so cannot reach a mature exon–exon junction. A parent duplex of 11 or 12 contiguous base pairs that pairs the whole catalytic gap is therefore invisible to all three, while being exactly what RNase-H1 requires.

Screen 4 compares every design's target window to every window of all six mature parents. 87 of 190 designs have a duplex of at least ten base pairs, and 61 of those 87 are against wild-type *NR4A3*, the transcript this modality must spare. The count is a floor at the threshold chosen, and seven base pairs is the strict end of the same cited range (§6): at seven the same screen returns 175 of 190 rather than 87. Those 61 are not 61 distinct sites: 59 of them are the same one, the mature exon-2/exon-3 seam every design's acceptor half reaches, which is the mature-transcript counterpart of the pre-mRNA concentration above. A 62nd pairs *NR4A3* at eleven base pairs but another parent at twelve, so it is attributed elsewhere. The count falls steeply with the gap-level margin: 50 of 76 designs at margin 1, 29 of 76 at margin 2, and 8 of 38 at margin 3. That is what the margin's definition predicts, since at margin 1 a parent needs one lucky base to pair the whole gap and at margin 3 it needs three. Five of the nine designs of §2.4 carry such a duplex at 11 or 12 base pairs, including 5′-CAGGGCATATCTTGCA-3′ against wild-type *NR4A3* itself. The margin is therefore a predictor of parent engagement rather than a guarantee against it, because it counts bases unique to the fusion at the junction without asking whether a parent carries them elsewhere.

Eighty-seven of 190 is 45.8%, with a nominal binomial Wilson 95% interval of 38.9–52.9% — nominal because it treats the 190 records as independent draws, which the close of this section states they are not, so it is narrower than a junction-clustered interval on the same counts. A count of that kind means little without a null, so the same screen was run over arbitrary 16-mers. Only the query changes: the same six mature parents, the same forward orientation, the same ten-base-pair threshold. Scrambling each design's own target window, which preserves its base composition and is the scrambled-gapmer control §4.4 asks a laboratory to make, gives 6.2% (5.9–6.4%); a dinucleotide-preserving shuffle gives 10.0% (9.7–10.3%); 16-mers drawn from uniform bases give 6.9% (6.7–7.2%), and from the panel's pooled base composition 7.2% (6.9–7.4%). A calculation agrees with the sampled figure rather than the sampled figure standing alone: the gap must pair, at 4^{-6}, and the run must then extend four further nucleotides across the two wings, at 1/64, which over the 19,921 parent windows searched predicts 7.3%. The observed rate is about sevenfold that, and the arm the modality actually turns on separates further still: 32.1% of designs pair the gap against wild-type *NR4A3* specifically, against 1.8% of scrambles. Scrambles are the weakest null run here; against the structural chimera null below, the strictest, that same arm separates 32.1% from 9.3%.

A second null asks whether that excess is a fact about reported breakpoints or merely about the design rule. Joining a random window of a real donor parent to a random window of real *NR4A3*, split at a junction offset the panel's registers use, reproduces everything the rule specifies, namely donor sequence 5′, *NR4A3* sequence 3′ and the junction inside the catalytic gap, while destroying only the fact that the two pieces meet where a tumour joins them. Those chimeras reach 23.8% (23.3–24.2%). At the ten-base-pair threshold, therefore, about half the observed liability is generic to any chimera of these two transcripts, and about half is specific to the real junctions. That apportionment is a property of the threshold rather than a stable share: across the seven-to-ten window the threshold comes from, the generic half runs from 52% at ten to 94% at seven, so at the loose end of the cited range almost all of the liability is what any parent-to-parent chimera gives. Two further arms locate it no more finely: holding the six gap bases and scrambling the wings gives 9.1%, and the mirror gives 8.8%, because a run reaching ten base pairs needs the real gap and the real flanks together. None of these rates is a significance test and none is offered as one. The 190 records are 176 distinct molecules tiled at overlapping registers across 38 junctions, so they are not independent draws, and a test treating them as 190 would be wrong about its own denominator.

2.6 · The *NR4A3* exon-2 acceptors, and the un-rearranged allele

Every junction above joins a donor exon to *NR4A3* exon 3, because the atlas that emits them drops exon-2 acceptors as non-coding: *NR4A3* exon 2 lies upstream of the start codon, so a junction at that acceptor cannot be built from the 38-junction panel at all. A frame-based exclusion is correct for a fusion protein and does not transfer to an RNase-H mechanism, which cleaves a transcript whether or not its reading frame survives, and the acceptors it excludes are not hypothetical: the *EWSR1* type 2 transcript joins *EWSR1* exon 7 to *NR4A3* exon 2,[24] sequenced as one of the five cases of a whole-transcriptome cohort[25] and again in an independent patient,[26] while functional work uses the *TAF15* exon 6 to *NR4A3* intron 2 variant.[27]

Those acceptors are now designed and screened to the panel's depth, at four seams with a published exon-resolved breakpoint, each tiled by the same five registers and graded by the same five screens. Their best available designs are 5′-CAGTGGGCTTCTGCTG-3′ at *EWSR1* exon 7, the type 2 transcript, at gap-level margin 2 and 51 gap-paired near-matches over 7 loci; 5′-AGTGGGCTCTCCACGG-3′ at *EWSR1* exon 13, at margin 3 and 25 over 6; 5′-AGTGGGCTCTTGTGTG-3′ at *TAF15* exon 6, at margin 3 and 128 over 6; and 5′-AGTGGGCTCTTCCATT-3′ at *PGR* exon 2, a seam reported in a single patient,[28] at margin 3 and 51 over 14 (Table 7). None of the four is clean, and the least loaded is the least dirty of five dirty designs rather than a different kind of result. They are reported beside the panel and never pooled into it, because the grade that excludes their junctions from the 38 is unchanged. *PGR* carries a further caveat: it is a sixth partner, outside the five modelled here, and §6 lists no transcript accession for it, so its design is screened against the non-canonical-acceptor table rather than derived through the panel's own transcript models.

Some designs cleave the patient's own un-rearranged *NR4A3* allele, and the parent screen passes every one of them. This is the result most consequential for anyone ordering these oligonucleotides, and it applies wherever a design's acceptor half is *NR4A3* sequence that is not exonic in the mature transcript: the 5′ untranslated exon 2, and the cryptic exon within intron 2. In a fusion transcript that sequence follows the partner's donor exon; in the un-rearranged allele it follows *NR4A3* intronic sequence, so the question is whether the design's donor half matches that intron closely enough for the whole catalytic gap to pair. For three designs it does: at *EWSR1* exon 13 joined to *NR4A3* exon 2, 5′-CAGTGGGCTCTCCACG-3′ and 5′-GCAGTGGGCTCTCCAC-3′, each pairing across the wild-type intron-1/exon-2 boundary at two mismatches with neither inside the gap; and at *TAF15* exon 6 joined to the intron-2 cryptic exon, 5′-TGATGAGGGCCTTGTG-3′, likewise gap-paired at two mismatches. All three are named here as not to be carried forward and are excluded from every best-design field above. Both seams keep a reagent, 5′-AGTGGGCTCTCCACGG-3′ and 5′-ATGAGGGCCTTGTGTG-3′, the second's catalytic gap carrying three *TAF15*-derived bases the *NR4A3* locus does not have and returning no wild-type site at all. What is lost is not a seam but the assumption that designs tiled across one seam are interchangeable.

Two things about that finding matter more than the three sequences. The first is how they were reached. Each had already cleared the mature-parent exclusion, and so had every other design at its seam, that exclusion being a screen over spliced cDNA and structurally unable to see intronic sequence: a clean parent screen at such a seam is the silence of an instrument that cannot look at the compartment in question. The same three were returned independently by an exhaustive scan of the *NR4A3* unspliced sequence, by the pre-mRNA screen and by the genome scan, on a fixed known-positive control that fired on exactly the one design it was required to fire on. The second is what decides it, because the obvious answer is wrong. It is not the gap-level margin this paper ranks by: the two condemned *EWSR1* designs carry 2 and 1 donor bases inside the catalytic gap, while a design at the same seam with a margin of 1 and five donor bases in its gap is clean. What decides is how much donor sequence the gap holds, the rest being acceptor sequence the wild-type allele already carries verbatim — a property of the donor rather than of the acceptor. The sequence bears it out: *EWSR1* exon 13 ends CACTCCGTGGAG against the last twelve nucleotides of *NR4A3* intron 1, CCTTGCCTGTAG, matching at 7 of 12 positions with a shared terminal AG, whereas *TAF15* exon 6 ends ACCACACACAAG and matches at 4, mismatching in every register — which is why the *TAF15* exon-2 seam returns no such design and the *EWSR1* exon-13 seam returns two. A design must therefore be checked against the acceptor gene's unspliced sequence whenever its acceptor half is not exonic in the mature transcript, and no spliced-transcript screen substitutes for it.

The *TAF15* intron-2 acceptor and a second cryptic-exon seam, *EWSR1* exon 10 to the same acceptor, are designed but not comparably screened: three of the five instruments cannot address them at all, resolving a junction by exon index where a cryptic exon 5′ of the *NR4A3* start codon has none, so their counts are absent rather than low and must not be read beside the panel's. The pre-mRNA and genome arms do reach them, and at the *TAF15* seam decide between its five designs rather than ranking them.

2.7 · The surviving candidates, and a genome-wide check

Composing the un-rearranged-allele exclusions of §2.6 with the deeper re-screen of §2.4 leaves three candidates in the whole panel, and they are not equally secure. 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8 and 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 carry no sense-strand near-match at ten times the default search depth, no single-mismatch match on the exhaustive transcript scan, no pre-mRNA site and no mature-parent duplex. Neither depends on the ten-base-pair threshold, because no window of any parent pairs their catalytic gap at any length: their longest run is zero rather than merely short. 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7 passes the same screens, but not in the same way. Its longest parent run is eight base pairs against wild-type *NR4A3*, which is below the threshold rather than absent, so it is a candidate at the stated cut and not at a stricter one. The fourth design with a clean deep screen, 5′-GGCATATCAAGCGCTG-3′ at the same junction, is excluded by an eleven-base-pair duplex with wild-type *TCF12* — its own donor parent, not the acceptor. That is the honest size of the candidate set, and no junction among them has a published patient breakpoint — the exon-resolved *TAF15* breakpoints reported in EMC are exon 6, and for *FUS* and *TCF12* exon 7 none has been published at all. Selecting within each junction rather than across the panel changes what is available, not what is clean: Table 4 applies the same criteria at all 38, where 35 have a design that clears the parent screen and *TAF15* exon 14, *TCF12* exon 3 and *TFG* exon 2 have none. All five junctions of the panel with a published exon-resolved breakpoint have one: four of them at the top gap-level margin, with longest parent runs of eight, eight, nine and seven base pairs, and *TFG* exon 7 at a margin of two with a longest parent run of nine.

Both classes were bounded the same way: exhaustive over six parent transcripts and silent about every other gene. The genome scan, screen 5, removes that bound.

A raw genome-wide count is not a result at this threshold. Chance alone predicts of order 10^3 near-matches per 16-mer over a genome for any 16-mer whatever, so the informative readings are stratified. Exact 16/16 matches are the class where chance expectation is of order one: 1.37 expected per design against 236 observed across 176 windows, which is at chance. Load relative to that expectation separates designs where a total cannot — the median design sits at 0.98 of its expectation and 14 of 176 exceed twice it. And the repeat split, free from a soft-masked reference, shows 52.5% of hits fully repeat-masked against a genome that is 51.4% masked, so the load is not repeat-driven.

The decisive reading is a lookup rather than a count: does any design have a gap-paired, strand-agreeing site in *NR4A3*, in a parent gene, or in an *NR4A* paralogue anywhere in the genome? Twenty of 176 do. No candidate above is among them, and the two secure at any parent-duplex threshold carry a load well below chance — 0.33 and 0.24 of expectation at ≤2 mismatches, and 0.06 and 0.04 for gap-paired sites, ranking 26th and 13th of 176. That is the strongest statement this work can make about them, and it is a statement about predicted hybridisation and not about cleavage.

2.8 · Expression of the off-target loci

No screen above establishes that a design's off-target gene is transcribed where the drug goes, and that discount applies to every count in this paper. Read against reference expression data (Table 6), the gap-paired loci of the best design at each of the four junctions covered separate in the direction opposite to the sizes of their loads. Table 6 lists every locus the deeper screens return across the tiling registers read at a junction, so its rows are a union over registers and can outnumber the per-reagent counts below, which are each best design's own. The *EWSR1* exon 12 reagent's six loci carry 123 of the panel's 649 transcript records, and none of the four measurable ones reaches the upper cut in liver or either kidney compartment; *ANKS1B* supplies 67 of them and sits below the lower cut in all three, peaking instead in brain at 24.9 TPM — GTEx's cervical spinal cord, a whole-body reading carried in the released artefact rather than in either of Table 6's two compartments, and *CHST5*, the smallest of the six by record count, peaks in gut on the same reading. The *EWSR1* exon 13 reagent's two loci are both transcribed at the upper cut in those same compartments. The *TAF15* exon 6 reagent's five loci separate the other way: *NRP1* reaches 6.6 to 17.8 TPM across all three exposure tissues and is the only one all five of that junction's tiling registers return, on five transcript records. It is at or above the upper cut in two of those three, and robustness to register orders the loci differently again; the tumour-compartment proxy orders them a third way. What these readings can and cannot decide between two reagents is stated where that choice is made (§4.1).

2.9 · Gap length trades junction specificity against parent-duplex competence

The panel above is one geometry. Tiling the same junctions at 5-8-5 and 5-10-5, wing fixed at five nucleotides, resolves what a longer catalytic gap buys and what it costs (Table 5, Figure 2).

What a longer gap buys and what it costs are the same nucleotide. Inside the gap, the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap. That holds for every design in all three panels rather than on average, and it fixes the direction of both trades. Within one geometry the gap is fixed, so the two move inversely: at 5-6-5 the 38 designs at margin 3 concede three nucleotides of contiguous parent-paired gap DNA, the 76 at margin 2 concede four and the 76 at margin 1 concede five. What the identity forbids is raising the margin past a geometry's ceiling, which is half its gap rounded down; that takes a longer gap, and a longer gap raises the parent-paired run at every register, the best-margin design running from margin 3 conceding 3 nucleotides at 5-6-5 to margin 5 conceding 5 at 5-10-5. Lengthening the gap therefore makes RNase-H1 more competent against the fusion and against the parent together, and no register choice within a geometry escapes that, though it does trade the two against each other.

Both directions are large. The best available gap-level margin rises from 3 to 4 to 5, and the junction-spanning registers per junction from five to seven to nine. At the *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 junction, the design carrying that margin sheds its transcriptome load completely: 123 sense-strand cleavage risks across the gap at six gene loci become 3 at one locus and then none. Over the six junctions screened at every geometry, designs carrying no such risk rise from 8 of 30 to 28 of 42 to 54 of 54, and the most risk loci on any one design falls from seven to two to none.

Against that, the contiguous DNA a wild-type parent pairs at the same junction rises from 3 to 4 to 5 nucleotides, and the most stable parent duplex from −7.77 to −8.66 to −10.25 kcal/mol. The corpus shows the same trade. Designs whose parent pairs at least five nucleotides of contiguous gap DNA, the shorter of the two reported minima for RNase-H1, rise from 76 of 190 to 228 of 266 to 342 of 342, and the median most stable parent duplex falls from −8.66 to −14.58 kcal/mol. At 5-10-5 that count is every design, and necessarily so, since the larger half of a gap of ten cannot be under five. At 5-6-5, 114 of 190 designs keep the parent below it.

Part of the fall in near-matches is guaranteed by the instrument rather than measured. At a fixed budget of two mismatches, every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the reachable set can only shrink as the design lengthens. Two mismatches is also a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall, and which designs reach zero, are measurements. The parent-side quantities carry no such qualification, being computed from the junction rather than from a search.

Two liabilities the transcript screens do not reach appear to move the favourable way, and neither reading survives one fixed criterion. A mature parent can pair the whole gap for 181 of 190 designs at 5-6-5, 130 of 266 at 5-8-5 and 87 of 342 at 5-10-5, but the whole gap is a six-nucleotide coincidence in the first and a ten-nucleotide one in the last, so the three counts are not the same test. Held to the ten-base-pair criterion applied everywhere else here, the liability is flat: 87 of 190, 88 of 266 and 87 of 342, the two criteria coinciding at a gap of ten because the gap alone is then already a ten-base-pair hybrid. Designs pairing the gap in parent pre-mRNA fall from 19 of 190 to 9 of 342, but that arm is a search at a fixed two-mismatch budget and inherits the nesting bound above rather than the parent-side quantities' freedom from it. Nor is the effect confined to the longest geometry: 5′-CAGGGCATATCAAGCGCT-3′ at *TCF12* exon 7 returns no near-match at all, where the 16-mer surviving at that junction returns three.

2.10 · Duplex thermodynamics and conventional design rules

Scored as free energies, every one of the 190 designs favours the fusion duplex over the best duplex either parent can form, by 4.8 to 13.1 kcal/mol with a median of 9.6. Every design favours the fusion because a parent pairs roughly half the oligonucleotide, and half a duplex is much the weaker one. That separates two things a base count conflates. Discrimination at the level of *binding* is not marginal here, and is not what constrains the modality. What remains unresolved is discrimination at the level of *catalysis*, where RNase-H1 requires a paired DNA gap and where the literature bounds span one- to five-fold. The thermodynamic result therefore narrows the paper's central uncertainty rather than relieving it.

The two rankings agree in direction. Grouping designs by the gap-level margin the Methods (§6) define, mean ΔΔG°37 rises monotonically with it, from 8.3 kcal/mol at margin 1 to 9.9 at margin 2 and 10.7 at margin 3. That agreement is arithmetic rather than corroboration: the longest run either parent can pair is exactly 11 minus the gap-level margin for all 190 designs, so the free energy is ordering the same quantity in kilocalories. What it adds is the size of the difference, not an independent ranking, and the same caution applies to the margin's agreement with the parent screens of §2.5.

Conventional design rules select differently, and against the paper's own candidates. Of the 190 designs, 106 satisfy all four rules; the rules bind at different rates, with every design free of a G-quadruplex motif but 13 carrying a homopolymer run of four, 43 a CpG dinucleotide and 58 falling outside the 40–60% GC window. The failures overlap, so they do not sum to the 84 designs that fail at least one.

The disagreement is sharpest where it matters most. Of the nine designs with no sense-strand near-match (Table 3), exactly one satisfies all four rules. Seven contain a CpG dinucleotide, the canonical TLR9 immunostimulatory motif; four fall outside the 40–60% GC window — the three *EWSR1* exon-1 designs above it at 62.5% and 5′-GGGCATATCTCTATAA-3′ below it at 37.5%. Only 5′-CAGGGCATATCTTGCA-3′ at *TCF12* exon 9 passes every rule, and the multi-partner candidate 5′-GGGCATATCATCAAAC-3′, which is not among the nine, also passes all four. The cleanest designs this work found are, with one exception, molecules conventional triage would flag, in six of the seven cases for a CpG a base substitution removes. Both are reported rather than composed into a single score.

3 · Discussion

Designability is not the constraint. Junction-spanning designs exist at every in-frame NR4A3 fusion junction, though at three of them every design pairs a wild-type parent through the catalytic gap. Nor does specificity sort by partner. With all 38 junctions screened, four of the five partners have a junction whose best design carries no sense-strand near-match across the gap at the deeper ceiling, and all five do at the default one. It is therefore the exon a fusion breaks at, not the gene it breaks into, that predicts a clean design — and the count of clean junctions is itself a function of search depth, as §5 sets out.

Clean designs are much scarcer than the default search depth implies, and two independent findings converge on that. One is search depth, measured in §2.4 and bounded corpus-wide in §5. The other is invisible to depth at any setting: five of the nine designs form an eleven- or twelve-base-pair duplex with a mature wild-type parent that pairs the whole catalytic gap, one of them with *NR4A3* itself, where no screen filtering on global identity can see it. Three designs survive every screen applied here, two of them at any parent-duplex threshold (§2.7).

The limiting step is discrimination between the fusion and its parents, and it is not resolved here. Both cited bounds are measured against a single substitution in an otherwise fully paired duplex. Neither transfers to a parent that leaves half the oligonucleotide unpaired and the catalytic gap only partly so: they bound the near-match case, and no retrieved measurement bounds the parent case. The two parent compartments of §2.5 sharpen that rather than softening it. For nine designs the route to wild-type *NR4A3* is not a gap-level discrimination problem at all. They pair the catalytic gap in full across the wild-type intron-2/exon-3 boundary, at two mismatches that both fall in the LNA wing, and the compartment in which that duplex would form is the nuclear one RNase-H1 occupies. For 87 the same is true in mature parent sequence, over a contiguous duplex of at least ten base pairs. The general point is that a fusion-junction design's most plausible wild-type liability is its own parent, reached either across a splice junction or in the mature transcript, and both are invisible to a screen that ranks candidates by global identity. A third compartment is invisible to all of them: at a seam whose acceptor half is not exonic in the mature transcript, the patient's own un-rearranged *NR4A3* allele carries the same sequence behind an intron, and three designs every parent screen passed pair their whole catalytic gap there (§2.6). The four demonstrations of parental sparing cited here were made in cells, on molecules already synthesised; the comparison above is available before anything is synthesised. Whether other groups apply an equivalent comparison before synthesising is not established here: no survey of published design pipelines was performed, and the argument above is about what particular instruments can see rather than about what the field does. The null of §2.5 is what makes that a finding rather than a restatement of the design rule: arbitrary sequence meets this screen at about 6%, and a chimera keeping the whole rule while joining the two parents at random offsets meets it at 24%, against 46% for designs at real breakpoints. At the ten-base-pair threshold, roughly half the liability is therefore inherent in joining these two transcripts at all, and roughly half is specific to where the disease joins them — a split that holds at that threshold and not across it, the generic share reaching 94% at the loose end of the cited window (§2.5).

All of that presumes that sparing wild-type *NR4A3* is worth the specificity cost, and that premise deserves examination rather than assumption. The published evidence cuts both ways and neither way is decisive. On the permissive side, *NR4A3* has two close paralogues and the family is functionally redundant where it has been tested: NR4A1 and NR4A3 are described as functionally redundant suppressors of acute myeloid leukaemia, and the three receptors are highly homologous.[29,30] A conditional double knockout of *Nr4a1* and *Nr4a3* disturbs haematopoietic stem-cell homeostasis, and even then the cells retain regenerative and differentiation capacity;[29] no single-knockout arm is reported there, so the double deletion is shown to be sufficient and not to be necessary. On the restrictive side, a separate study makes the loss of *NR4A3* consequential rather than silent when paralogue reserve is reduced: mice hypoallelic across the two genes develop a myelodysplastic or myeloproliferative neoplasm, and abrogation of both leads to rapid postnatal leukaemia.[31] The family is also not uniform in direction — within atherosclerosis, NR4A1 and NR4A2 attenuate lesion formation while NR4A3 aggravates it[32] — so paralogue redundancy cannot be assumed to be substitution.

Two limits on that reading matter more than the reading itself. Every study cited here is haematopoietic or vascular, and none addresses the tissue an EMC arises in; and all describe germline or conditional gene deletion, which is a different and more complete perturbation than partial, reversible, dose-limited knockdown by an oligonucleotide. The honest position is therefore that wild-type *NR4A3* knockdown has an unquantified cost that is probably not zero and probably not catastrophic, and that the case for junction selectivity does not rest on it. It rests on the *EWSR1* and *TAF15* side: the fusion's partner genes are essential RNA-binding proteins, and a reagent cleaving a parent transcript is failing at the one thing that distinguishes this modality from knocking down *NR4A3* directly, which requires no junction at all. A design that cannot spare the parents has no advantage left to trade.

Free-energy calculation does not narrow the interval either. Every design discriminates amply at the level of duplex formation, so what is unresolved is specifically the catalytic step, not the binding one. Two things could narrow that interval, and no further sequence analysis is either of them: a measurement, or a physics-based estimate of cleavage geometry on the RNase-H1·heteroduplex complex, for which experimental structures exist. Neither is attempted here. Gap length is not a third, for the arithmetic reason §2.9 gives: a longer gap buys a markedly quieter transcriptome by making RNase-H1 more competent against the parent as well as against the fusion, which is the same limit reached from the other side rather than a way around it. The field's own answer to poor single-base discrimination has been positional chemical modification of the gap rather than length,[33] and that is the design direction this result points to, now for a demonstrated reason rather than by analogy. A steric-block mechanism, which does not require gap-level discrimination, is a second alternative this work does not evaluate.

Delivery remains unsolved for a tumour, and separates into three routes with different requirements. A characterised EMC-enriched surface antigen is a prerequisite of the systemic receptor-targeted route only; local and inhaled administration require none. EMC's distant spread is lung-dominant, at 35–45% of patients and a median of approximately 28 months to metastasis.[6] Inhaled oligonucleotides have reached human dosing in non-oncology indications. An inhaled antisense oligonucleotide has been dosed in healthy volunteers in phase 1,[34] though that was a splice-switching oligonucleotide rather than an RNase-H1-active gapmer, so it establishes the route and not the mechanism used here. An inhaled siRNA has reached phase 2b–3 in patients.[35] Both target airway epithelium or parenchyma, which is the compartment inhalation naturally reaches. A hypocellular, matrix-rich parenchymal sarcoma nodule is not. Inhaled delivery to lung tumours is an active preclinical field, with 68 records in the retrieval corpus behind this section, but only two of those carry clinical-stage language and neither is a trial. The route is therefore established in humans and not for this target.

The testable surface is narrower than the literature makes it look, and a reader planning an experiment should know that before ordering anything. A junction-spanning gapmer needs a junction to span, so the test article has to carry one. The EMC line a reader would reach for first, H-EMC-SS, is the only one this work could establish as available from a cell repository, a third line's distributor being unreadable and its availability therefore unanswered rather than answered, and no *NR4A3* fusion is detectable in it on the public record: a filtered fusion caller that ran against it returned two calls, neither naming *NR4A3* nor any FET gene; its *NR4A3* expression sits at floor; the reference registry that records a gene fusion for other EMC models records none for this one; and no retrieved source reports a positive junction in it. The operative consequence is narrow and is the only one this paper draws: no reagent named here can be tested in that line. This is not a statement that the line is misidentified — it carries a concordant short tandem repeat profile from three independent sources and no problematic-line flag — nor a statement about what the line is instead, and fusion-negative EMC tumours are themselves a recognised minority category, so absence of the fusion is not by itself a reclassification. The observation is also not new: it is in print in one figure legend and is carried as a caution field in the reference registry. No retrieved source examines it as a subject, and it is not discoverable by anyone searching on model validity.

What remains is five test articles, and each of the five now has a matching reagent. Three are the engineered constructs of the functional study cited above,[27] E-N, T-N* and T-N, whose exon spans that paper states verbatim; two of the three are junctions this work screened at full panel depth and the third is the intron-2 cryptic-exon seam of §2.6. The other two are the patient-derived, identity-clean models reported with two EMC tumours,[36] whose fusions are reported as *EWSR1* exon 13 and *TAF15* exon 6 joined to *NR4A3* exon 2 rather than exon 3; reagents exist at both acceptors, so each line has one under either reading of that exon label.

The two routes are not interchangeable and their limits run in opposite directions. Rebuilding the constructs is the faster route and the only one whose critical path contains no third party who can decline or not reply, since the junction is specified by construction. What it cannot buy is biological relevance: a complementary DNA over-expressed in a heterologous background is not the disease, so such an experiment could speak to junction-selective knockdown of the intended transcript and not to activity at endogenous expression from an endogenous locus. The published recipient background is in no cell-line registry either, so a rebuild would sit in a different background from the original — an isogenic mismatch to declare rather than gloss. The patient-derived models are the only route to a fusion-positive EMC cell, and are available on request from the originating laboratory with no repository deposit; what a transfer requires is stated nowhere that could be read, which is an absent statement rather than an absence of conditions, and the cells are slow once received, at reported doubling times of five to six days as sarco-spheres passaged every two to three weeks, which constrains any exposure window. A third reported line cannot serve as a test article at all on current evidence, because its fusion partner and exons are unstated anywhere readable and it would have to be sequenced first. One constraint sits above all of them and no reagent choice moves it: every route ends at someone culturing cells, and this work has no laboratory, so the rate-limiting step is a laboratory rather than a line, a construct or an oligonucleotide.

One deliverable does not wait on that. No named reagent reaches every patient, and the released design and screening pipeline is the paper's second output: §4.5 states what it takes as input, what it does with it and what its result is worth, which is a candidate rather than a validated reagent.

4 · Reagents, controls and the decisive experiment

This section is the paper's output for a laboratory. It names six things: the oligonucleotides to make, the arm that separates the two ways a weak result could arise, the predicted off-target load each carries, the controls without which the readout does not mean what it appears to mean, the number that would falsify the ranking every candidate here is ordered by, and — for the patients no named reagent reaches — the released procedure by which a reagent can be designed for a breakpoint outside this panel. Nothing in it is a claim of efficacy. No sequence named below has been synthesised or tested.

4.1 · The reagents to synthesise, and their population coverage

The experiment that would resolve the central uncertainty has been published in an analogous disease. Fusion-specific antisense oligonucleotides against *NAB2::STAT6* in solitary fibrous tumour, evaluated against CRISPR-engineered isogenic fusion-positive and fusion-negative cells, reduced fusion expression by 58% and proliferation by 22% in vitro.[37]

Applied here, the reagents to synthesise are the best available at the two most frequently reported junctions with a published exon-resolved breakpoint (Table 4): 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6. Both hold the top gap-level margin of 3, and neither pairs a parent through the catalytic gap at the ten-base-pair threshold, although the *TAF15* reagent's longest parent run is nine. The first also tests the multi-partner prediction, against a synthetic target only.

How much of the disease those two junctions represent is a junction figure, not a partner figure, and the two differ: 46 *EWSR1* and 9 *TAF15* among 58 molecularly confirmed cases is 95% between the partners,[9] while each reagent spans one exon pair. Discounted by the breakpoint distribution of an 18-case series[22] the two are 68.4%, roughly two thirds; Table 7 gives that figure, the rungs above it and the reagent at each. The interval is wide for the denominators rather than the estimate: taking each breakpoint fraction to its own Wilson bound spans 39.9% to 82.8%, the *EWSR1* arm resting on 15 tumours and the *TAF15* arm on three. It assumes too that the breakpoint distribution within *EWSR1*-rearranged tumours is the same in the 58-case cohort as in the 18-case one, which nothing here tests and which no published series is large enough to settle. Breakpoint sequencing of archival material would sharpen it; no further analysis of sequence will.

Two things sit outside the table. Coverage rises only by adding reagents, because no oligonucleotide can serve two breakpoints of the same partner: *EWSR1* exon 12 ending AATGGTTTGATG against exon 13's CACTCCGTGGAG, which agree over a single terminal base. Taken across every pair of in-frame junctions of one partner in this panel, the longest shared 3′ donor run is five nucleotides, at *TFG* exons 2 and 6, and three within *EWSR1*. The cross-partner coverage of §2.2 is the exception that shows the rule, needing the ten identical donor bases the FET paralogues share. And the *NR4A3* exon-2 acceptor rows of §2.6 sit in Table 7 without entering the panel's own counts.

Two further reagents extend the set, both at the top gap-level margin of 3: 5′-GGGCATATCTCCACGG-3′ at *EWSR1* exon 13 to *NR4A3* exon 3, and 5′-GGGCATATCCATCAGA-3′ at *TCF12* exon 5 to *NR4A3* exon 3, whose junction is resolved to the nucleotide by the deposited chimeric cDNA of §2.3. The second comes with no distribution: one *TCF12*-rearranged tumour has ever been sequenced at this junction, neither breakpoint series contains a *TCF12* tumour at all, and the break-apart assay the later cohorts used locates no seam within the *NR4A3* locus, so recurrence there is untested rather than refuted and the resulting 98.3% is an upper bound rather than a reachable target. SI §S6 carries the rest of the ladder's bookkeeping.

The *EWSR1* exon-13 reagent should not be recommended on its transcriptome count, because the two axes that separate it from the exon-12 reagent point in opposite directions and only one of them bears on where an effect would land. On count it is the lighter of the two, 24 gap-paired near-matches at 2 loci against 123 at 6. On exposure it is the heavier: both of its loci are transcribed at the upper cut in the organs a systemic phosphorothioate gapmer distributes to, where none of the exon-12 reagent's measurable loci is (§2.8, Table 6). For a laboratory choosing between them, the exposure reading is the one that speaks to the question a count cannot: a locus matched but not transcribed where the drug goes has no route to an effect, whereas the count is a property of how densely the returned loci are annotated. Neither axis is a risk ranking, and this comparison does not make the exon-12 reagent the safer molecule. Every hit behind both is a 14 of 16 match, no cleavage is predicted at any of them, and an expressed gene is necessary and not sufficient for an effect. The reason to make the exon-13 reagent is coverage, and it is unaffected by either axis.

Two risks attach, in this order. The first is architectural, and the Methods (§6) disclose it. A six-nucleotide gap supports noteworthy but incomplete RNase-H1 activity where seven to ten are reported as optimal,[38] so weak knockdown is at least as likely to be the gap as the sequence. That risk is now addressable by a named second reagent rather than by a caveat.

4.2 · A second geometry as a gap-length control

5′-AGGGCATATCATCAAACC-3′ is the 5-8-5 design at the same *EWSR1* exon 12 junction. It spans the same three partners' breakpoints and sits inside the reported activity optimum. It holds a gap-level margin of 4 where the 16-mer holds 3, and carries 3 sense-strand near-matches across the gap at one gene locus, against the 16-mer's 123 at six (§2.9, Table 5).

Synthesised alongside the 16-mer, at one extra oligonucleotide and one extra well per condition, it separates the two explanations a weak result would otherwise confound. A 5-8-5 arm that knocks down where the 5-6-5 arm does not attributes the failure to gap length rather than to sequence. What it does not buy is parental sparing, since the same two nucleotides lengthen each parent's contiguous duplex from 3 to 4 nucleotides of gap DNA, its whole contiguous hybrid at that seam from 8 to 9 base pairs, and its free energy from −7.77 to −8.66 kcal/mol. Its longest mature-parent duplex through the whole gap does fall from 8 base pairs to none, but 8 sits below the ten-base-pair threshold applied throughout, so neither design counted as a mature-parent liability at this seam and that fall removes nothing the screens had counted. Both arms therefore need the fusion-negative comparator below.

4.3 · The predicted off-target load of each reagent

The second risk is transcriptome load, and it differs sharply between the two reagents. The *EWSR1* reagent carries the heavier load of the two named here: 123 gap-paired sense-strand near-matches at the deeper ceiling, recounting to six gene loci, all at the screen's loosest admitted identity and none on a parent transcript (§5). It is not the heaviest in the panel — fourteen of the 187 design records re-screened at that ceiling carry more, to a maximum of 240 at *TCF12* exon 3. Table 2 prints one row per junction rather than one per design record, so those per-record counts are in the released screens rather than in a table. The *TAF15* reagent carries 8 such near-matches at five loci.

The parent compartments qualify that, in a way the transcript screens cannot show. The *EWSR1* reagent carries a sense-strand intron–exon-spanning near-match in wild-type *TAF15* pre-mRNA at two mismatches, one of them inside the catalytic gap, returned independently by the pre-mRNA screen and the genome scan. It falls outside every parent count reported here, because those require the gap to be paired in full, and by the bounds adopted above a single gap mismatch does not abolish cleavage. It is the multi-partner result's own cost rather than an incidental hit:

the ten donor bases shared across *EWSR1*, *TAF15* and *FUS* that let one oligonucleotide span three junctions are the bases that place it against wild-type *TAF15*. The *TAF15* exon-6 reagent carries no sense-strand pre-mRNA site at all, which is a second respect in which the two separate on something other than count.

That load should travel with the reagent. It is a liability to disclose and to control for rather than a disqualification, because on the genome scan the same design falls below chance in both directions that matter: 0.69 times the expected number of near-matches at two mismatches, and 0.62 times the expected number of gap-paired ones. Expression separates the two reagents the other way (§2.8, Table 6): none of the *EWSR1* reagent's measurable loci is expressed at the upper cut in the organs a systemic dose reaches, while the *TAF15* reagent's five include *NRP1*, which is. That does not reverse the ranking, since no screen here establishes that a two-mismatch duplex engages any of them, and it is not a statement about safety.

4.4 · Controls and the decision threshold

The three designs that survive every screen are mechanism controls rather than candidates: 5′-AGGGCATATCGGAGTC-3′ at *FUS* exon 8, 5′-GGGCATATCCGACATG-3′ at *TAF15* exon 1 and 5′-GGGCATATCAAGCGCT-3′ at *TCF12* exon 7, tiered as §2.7 describes. None sits at a junction a patient is reported to carry (§2.7), which is what makes them controls rather than candidates. 5′-GGGCATATCTCTATAA-3′ at *TCF12* exon 17, which the default-depth screen returns as clean (Table 3), is not among them. At ten times the default search depth it carries 101 sense-strand near-matches, 14 of them spanning the catalytic gap.

What a knockdown experiment with these reagents can transfer to depends on how it is set up, and the first requirement is upstream of the assay. The breakpoint of the cell line or patient sample must be established at nucleotide resolution by RNA sequencing before any oligonucleotide is ordered: every design here is specific to one exon pair, and none is valid for an unverified junction. Routine diagnosis does not supply it: break-apart *NR4A3* fluorescence in situ hybridisation is the preferred single assay because it detects any rearrangement "irrespective of partner",[6] so on its own it does not locate the seam, and the 58-case cohort every coverage figure here is denominated on was identified that way, with RNA exome sequencing applied to three of its cases.[9] That cohort reports a partner for 57 of its 58 tumours, so partner assignment there rests on more than the break-apart assay; what no part of that workflow supplies is the exon pair.

Three controls are required, and a knockdown assay alone distinguishes none of them:

- a positive control gapmer against an abundant housekeeping transcript in the same cells, to separate failed delivery from failed discrimination;
- a scrambled gapmer of matched chemistry, to separate sequence-specific cleavage from the non-specific toxicity of this chemistry;
- a fusion-negative isogenic comparator, since wild-type *NR4A3* may be too weakly expressed in an EMC line for the selectivity readout to be defined at all.

Two limits of that set are invisible within it. The normalising reference transcript must be neither the positive control's target nor either transcript measured, or the positive-control well becomes uninterpretable and every other well is rescaled against a perturbed reference. And none of the three separates RNase-H1-dependent cleavage from steric block, which §3 names as unevaluated, so a knockdown seen with this set is consistent with both while the ranking under test models cleavage alone.

Where the wild-type measurement sits decides the answer, because the fusion carries *NR4A3* exons 3 to 8 and a wild-type assay must not read sequence the fusion also carries. The two available placements bias in opposite directions. An amplicon upstream of the acceptor cannot detect the fusion, but it lies on the 5′ fragment that survives cleavage, so it under-reads wild-type knockdown and inflates apparent selectivity; an amplicon spanning the wild-type exon-2/exon-3 boundary reads knockdown directly, but at the *NR4A3* exon-2 acceptors of §2.6 the fusion itself carries that boundary. Both are defensible, no assay is prescribed here, and which placement was used should be reported with the result.

The decision threshold should be fixed before the experiment, in a form that can be registered. Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose–response in the same wells, so that a larger number is a more selective reagent. The cut is defined for that quantity and no other. A ratio of residual transcript at a single dose is not commensurate with it and must not be compared against the same cut: that ratio is bounded above by one divided by the fusion knockdown's complement, so at the 58% knockdown reported for the analogous published experiment[37] it cannot exceed about 2.4 however selective the reagent is, and a cut of 5 would return falsification as arithmetic rather than as biology.

The replicate count follows from the variance rather than being asserted. Selectivity here compounds four normalised measurements, so the replicate standard deviation of its logarithm is the quantity to estimate in a pilot. At a standard deviation of 0.35 on that scale, six independent biological replicates give about 80% power to falsify a true selectivity of 3, and three give about 30%; above a standard deviation of about 0.65 no observed ratio can place a 95% upper bound below 5 at three replicates, so the test cannot fail at all and the design is void rather than negative. The number of replicates should therefore be set from the pilot estimate, with three as a floor and not a target, and the ranking is falsified only where the upper bound of the 95% confidence interval lies below the cut, never where a point estimate does. The ratio is reportable only where the wild-type *NR4A3* knockdown is itself resolved above a pre-stated limit of quantification — the quantity that can approach zero, which is the change in wild-type transcript and not its vehicle-well abundance; a wild-type change indistinguishable from none returns an unbounded selectivity that would otherwise pass by default, and the honest output there is a one-sided lower bound, which cannot falsify. The cut is 5.0, taken as a convention from the approximately five-fold near-match figure of §6 rather than measured for this comparison, on which §3 records that no retrieved measurement bounds the parent case. It sits at the optimistic end of the one- to five-fold span reported above, so a reading inside that span is consistent both with the ranking failing and with the discrimination this chemistry is already reported to give; the margin contrast below is what separates those two, and a single-arm result does not.

The reagents of §4.1 all hold gap-level margin 3, so as a set they measure the level of selectivity at the top margin rather than whether selectivity orders by margin. One optional arm supplies that contrast, and only at *EWSR1* exon 12: 5′-GCATATCATCAAACCA-3′ is the margin-1 register of the same junction at the same geometry. Margin is not the only variable that moves with the register, and the others are named here rather than left for a reader to find: GC falls from 43.8% to 37.5%, outside the 40–60% window §2.10 audits, so this arm would fail a conventional rule the lead reagent passes; gap-paired near-matches fall from 123 to 34 while single-mismatch off-targets rise from 1 to 22; and its longest mature-parent duplex is eight base pairs against wild-type *FUS*, which is the same length the lead reagent carries against wild-type *TFG*, so the two do not separate on that screen at all. A difference between the arms is therefore not attributable to margin alone.

One asymmetry decides what the arm can show. Its five parent-paired gap nucleotides are *EWSR1*, the donor, not *NR4A3* — so the liability its low margin creates is invisible to a readout defined on wild-type *NR4A3*, and interpreting this arm requires the wild-type *EWSR1* measurement alongside it. The mirror register at the same seam, 5′-CAGGGCATATCATCAA-3′, is the one whose five fall on *NR4A3*, and it carries an eleven-base-pair duplex against wild-type *NR4A3* that would confound it differently.

TAF15 exon 6 cannot supply the same arm: all four of its lower-margin registers pair a parent through the whole gap at eleven or twelve base pairs, two of them against wild-type *NR4A3*. A contrast at one junction tests the ordering at that seam and not the ranking across the panel, and the comparison it supports is between two arms rather than of one ratio against the cut, for which this section states no rule.

4.5 · A design procedure for a breakpoint outside this panel

The reagents named above do not reach every patient: the two leads cover roughly two thirds of molecularly confirmed cases, and the panel is bounded by what has been sequenced rather than by what can be designed. The deliverable is therefore the procedure as well as the reagents, and it is the procedure that produced this paper's 190 designs, released unchanged with the artefacts.

Its input is the breakpoint at nucleotide resolution, by RNA sequencing of the tumour, which §4.4 requires in any case; an exon pair inferred from a break-apart assay is not sufficient. Given a declared exon pair, `junction_aso.py` retrieves the parent transcripts, builds the modelled fusion, grades the pair for frame and tiles the junction-spanning gapmers, emitting each with its GC, its gap-level margin and a check that it complements neither parent perfectly. The five screens then run on that panel: `junction_aso_offtarget.py` the alignment screen against human RefSeq RNA, classifying each near-match by whether the catalytic gap is paired; `aso_insilico.py` the exhaustive transcript scan, the target-site accessibility fold and the sequence-liability filters; `aso_premrna_offtarget.py` the parents' unspliced sequence; `aso_genome_offtarget.py` every position of GRCh38; and `aso_parent_gap_pairing.py` the mature-parent screen. A new design must clear all five, and the parent screens matter most: pairing a parent through the whole catalytic gap is this paper's central negative and surrenders the only advantage the modality has. Where the acceptor half is not exonic in the mature transcript, the un-rearranged-allele scan of §2.6 applies as well.

What that yields is a candidate, not a validated reagent. Nothing designed this way has been synthesised or tested, the procedure has been run at the junctions reported here and at no others, and a design it returns is ordered by the ranking §4.4 states a falsification threshold for.

5 · Bounds on every claim, and the conditions for falsification

Seven bounds sit under everything above, ordered by how much a decision would change if one of them bit. None alters the ranking by gap-level margin; each alters what a count taken from it means.

Counts are lower bounds, and a zero is not exempt. The alignment screen is heuristic and stores at most 50 hits per query, so 136 of the 183 filtered designs carry right-censored counts: 35 at the cap, 101 more past the 15 hits retained. Re-screening at ten times the ceiling raised the count for 164 of the 180 designs screened at both depths, 129 of which had never approached the cap, so reaching the cap is not what censors a count. Depth then moves the headline result: six of the nine clean designs lose the property at the deeper ceiling, three having returned no near-match at all at the default one (§2.4). Only 47 of the 183 hit lists are short enough to assess for cleanliness, so nine is a floor over that subset and an over-count of it at once, and their own raw counts, zero to eight hits each, are not a measurement either. Retention alone withheld a verdict on seven further records — a different seven from the seven whose default-depth query failed at the remote service (§2.4) — and the deeper pass decided six of the seven; none of the six is clean, and the seventh re-screen did not return, so that record remains undecided. The nine are untouched by that test. BLAST's sensitivity at ≥14/16 is unquantified here, so "no sense-strand near-match" is a property of this search and not of the transcriptome; the exhaustive transcript scan, complete for substitutions by construction, is the screen the claim rests on.

The parent threshold is stated, not measured. Every parent count — 87 of 190, 61 of them against wild-type *NR4A3* — is taken at a contiguous duplex of ten base pairs, a criterion adopted here rather than measured for this architecture, so the count is a floor at that choice: at the seven-base-pair end of the same cited range the screen returns 175 of 190. Nor does the criterion transfer cleanly, ten being a whole-duplex length where the source counts RNA:DNA nucleotides, which this architecture holds at six (§6). What it apportions moves with it: the generic share of parent liability runs from 52% at ten base pairs to 94% at seven (§2.5).

The patient may not carry these junctions. Which exon pair a patient carries is not decidable from exon structure. Coverage prices *TAF15* at 3 of 3 on a three-tumour series, while a functional study reports a second isoform at a cryptic exon within *NR4A3* intron 2 and calls the two "the two major *TAF15-NR4A3* isoforms detected in human tumors" without counting either,[27] so that arm's coverage is an upper bound and 68.4% is optimistic on it by an unmeasured amount; no design in the panel crosses that seam, 0 of 190, and the two isoforms share no sequence 3′ of the breakpoint, so it is a different target and not a near-miss. The multi-partner result is conditional on *TAF15* and *FUS* breaking at the homologous exons, which is not established here, and the five partners are not the catalogue: *ACTB*[3] and others are reported, and 2% of one cohort carried no identified partner.[9] Three of the 15 *EWSR1*-rearranged tumours of the primary breakpoint series carry transcript types the retrieved record does not name, so retrieval is an upper bound on what would open that block. At least one named variant is reported to arise from a genomic breakpoint interior to *EWSR1* exon 12 rather than between two exons, in a source that carries a citation marker on that sentence and is therefore restating an earlier report.[39] Such a breakpoint is not undesignable: handing the same builder a donor model cut inside its exon returns five candidates, all fusion-specific against both parents, three gap-centred and a best gap-level margin of 3. What such a junction lacks is an exon index, which is how every design here is specified, and a published nucleotide position, which no retrieved source states, so it is out of reach for a stocked panel while remaining designable for a named patient whose breakpoint has been sequenced. How many tumours this accounts for is not established by any source.

One geometry. Every screened count outside §2.9 is for one architecture, a 16-mer at 5-6-5. The genome scan is unavailable at 18 and 20 nucleotides by construction rather than merely unrun, so the nesting bound on a longer design's genome liability is a next step and not a result, and no RNase-H1 assay distinguishes these geometries here.

Hybridisation, not cleavage, and not exposure. All five screens address hybridisation-dependent liability only, and the free-energy calculation speaks to duplex formation rather than to cleavage. Every parent count requires the catalytic gap paired in full, an inclusion criterion adopted because no retrieved measurement grades a partly-paired parent duplex; the class it excludes is the 21 designs of §2.5. Nothing here establishes that a matched gene is transcribed where the drug goes: 13 of the 46 loci returned no reading and carry 52 of the panel's 649 records, so there the exposure question is unanswered rather than answered negatively, and reference bulk medians describe a population's normal tissue rather than a dosed patient's organ (Table 6). The sequence-independent liabilities of this chemistry, protein binding and target-independent hepatotoxicity, are not a function of any feature graded here.

The chance null is crude. It assumes independent uniform bases, where real transcript sequence is composition-skewed and repetitive. An arbitrary position matches a given 16-mer at ≥14/16 with probability 2.6×10^{-7}, or 189 near-matches for any 16-mer whatever over the exhaustive scan's measured span — a figure the alignment screen's 50-hit cap cannot test at all. The scan itself can, and on the mean it comes in at chance: at ≤1 mismatch the same span predicts 8.2 per design against an observed 9.2 over the 176 distinct oligonucleotides, a ratio of 1.12. The shape disagrees in both tails, 40 of the 176 returning no match where a Poisson null of that mean predicts fewer than one, and a right tail reaching 100 matches on one design (Supplementary Figure S1), so the null separates "more than chance" from "at chance" and nothing finer and does not license reading the zeros as cleanliness.

Records are not genes, and one assembly is not a genome. RefSeq carries one accession per annotated variant, so a match to a constitutive exon is counted once per variant: over the 44 designs of the 38 junction screens whose hit lists permit a locus recount the median inflation is 2.25 records per locus and the maximum 11.0, more than doubling the apparent number of distinct genes. A design's *load* is its predicted off-target burden counted as near-matches, and where it sits matters as much as its size: `NM_`/`NR_` records are curated, `XM_`/`XR_` computationally predicted. Both compound on 5′-GGGCATATCATCAAAC-3′, whose 123 gap-paired near-matches at the deeper ceiling recount to six loci: 82 of the 123 are predicted models and the other 41 are curated records, and those 41 are themselves inflated, 32 of them *ANKS1B* accessions and three *ZNF667*. The curated share moves with depth as well. At the default ceiling that design carried a single curated sense-strand hit, *H2AP*, mismatched inside the catalytic gap and so counted in full under the pessimistic bound; at the deeper ceiling it carries 43. The genome scan decides sense-strand membership against an annotation, so unannotated transcription is uncounted; it is one assembly, silent on a patient's private variation; and every exhaustive screen here is complete for substitutions by construction and blind to insertions and deletions.

6 · Methods

Transcript models. Canonical transcripts for the five partner genes and for *NR4A3* were obtained from Ensembl.[40] Each model was self-checked before use: exon lengths must sum to the spliced cDNA, the CDS must occur exactly once within it, and translation of the CDS must reproduce the annotated protein. Per-exon coding content was additionally cross-checked against an independent exon audit for *EWSR1* and *NR4A3*; for the other four partners that audit does not exist, and the weaker check is recorded per gene in the released artefacts. Every exon number, coordinate and length in this paper is relative to one specific model per gene, and the canonical transcript of a gene can change between Ensembl releases, so the six accessions are given here rather than left to the artefacts: ENST00000397938 (*EWSR1*), ENST00000605844 (*TAF15*), ENST00000333725 (*TCF12*), ENST00000254108 (*FUS*), ENST00000240851 (*TFG*) and ENST00000395097 (*NR4A3*).

Chimera construction. Chimeras were built from transcript sequence rather than by joining coding sequences. A fusion keeps the whole *NR4A3* acceptor exon, so any bases of that exon lying ahead of the *NR4A3* start codon are still present in the fusion transcript, and they are the first bases an oligonucleotide meets on the *NR4A3* side of the junction. At the exon-3 acceptor, the only one that yields designs here, there are two. Joining coding sequences alone would omit them, shifting every design by two positions. A pair of exons is *in frame* when the partner's coding bases, plus those retained bases, sum to a multiple of three. Every declared exon pair was graded by that rule before any design was emitted, and only the in-frame pairs were carried forward, since only those describe a fusion that could exist.

Design. Junction-spanning 16-mer gapmers were tiled in a 5-6-5 LNA/DNA/LNA architecture on a phosphorothioate backbone, which is the chemistry the design rules below assume. Each way of sliding that 16-mer along the transcript is a *register*, and only registers placing the junction inside the six-nucleotide DNA gap were retained, since RNase-H1 cleaves within the DNA:RNA duplex of the gap and needs a minimum run of contiguous DNA to do so.

The gap length is a compromise and is treated as one. Reported minima for that run are five to six nucleotides — at least five for cleavage to occur,[41] or a DNA segment of "six or more bases" to activate the enzyme[42] — and for LNA/DNA/LNA gapmers specifically a six-nucleotide gap gives noteworthy but incomplete activity, with seven to ten reported as optimal.[38] None of the three figures is a titration in this architecture, and SI §S3 gives the provenance of each. Six therefore sits at the short end of the usable range and below the reported optimum. It was retained because it admits exactly five junction-spanning registers per junction. No claim is made that a short gap improves fusion-versus-parent discrimination: one series that shortened a 5-10-5 gapmer to 5-6-5 reported lower off-target knockdown but also lower on-target activity and lower allele selectivity.[41] Because that trade is the modality's central one, 5-8-5 and 5-10-5 were tiled over the same junctions by the same rule and carried through the same screens, wings held at five nucleotides so that only the gap changed and LNA affinity enters every parent duplex identically (§2.9).

Ranking. What separates the fusion from a parent is the junction-unique bases inside the gap, not identity across the whole oligonucleotide, because the gap is where the enzyme cuts. Designs were therefore ranked by their *gap-level margin*: the junction-unique bases inside the gap on the shorter side of the junction. That is the panel-level statistic, which compares designs across junctions. Selecting within one junction is a different question, and Table 4 — from which the reagents of §4.1 are taken — orders designs by parent liability first, then pre-mRNA sites, then distinct gene loci, with the margin breaking ties. Each candidate was screened against all six parent transcripts rather than the two of its own fusion, because the FET-family donors (FUS, EWSR1 and TAF15) are paralogues with similar low-complexity amino-termini.

Specificity screening. Five screens were applied. Each is named below and referred to by that name throughout, because each reaches a compartment the others cannot and each is blind to something another catches. No single screen supports any claim here on its own.

1. **The alignment screen.** Each target window was queried against human RefSeq RNA[43] with BLAST+[44] (blastn-short, low-complexity filter off, ≥14/16 identity). A transcript window matching a design at 14 or more of its 16 positions is a *near-match*, classified by whether the six-nucleotide gap is itself base-paired: one that pairs the gap is *gap-paired*, or gap-spanning, and RNase-H1 could cleave there; one pairing only the wings could not. This is a heuristic search retaining only a limited number of hits per query, an effect §5 measures. Records of the six parent genes are counted separately and excluded from every near-match count reported here, since each parent pairs one wing by construction and would otherwise dominate the list; the parents are assessed instead by the gap-level margin and by screen 4.

2. **The exhaustive transcript scan.** A seed-and-extend scan searched 186,185 transcripts (GRCh38.p14) for exact and ≤1-mismatch matches. It is complete for substitutions by construction, reads the sense orientation only, and does not detect insertions or deletions.

3. **The pre-mRNA screen.** Screens 1 and 2 search mature transcript, so a third covers the nuclear compartment they cannot reach. Unspliced sequence and exon coordinates for all six parents were retrieved from Ensembl, and every target window was scanned against them in both orientations at ≤2 mismatches. That is the threshold the alignment screen admits, which keeps the two comparable: a stricter one here would return a cleaner pre-mRNA result for that reason alone. This arm is exhaustive for substitutions, seeded on three blocks of the 16-mer so that a hit within the threshold must match one block exactly. Each hit is classified as wholly intronic, wholly exonic, or spanning an intron–exon boundary, since only the exonic class could have been visible to a transcript screen.

4. **The mature-parent screen.** This addresses the parent transcripts screen 1 excludes, in the compartment screen 3 cannot reach. Mature parent transcripts were spliced from the same Ensembl records, and every target window was compared to every 16-nucleotide window of all six, forward orientation only. A window counts only if all six gap positions are paired. Its size is the longest contiguous run of perfect pairing containing the whole gap, which is the duplex RNase-H1 would see. Runs shorter than ten base pairs are not treated as plausible substrates. That is a stated threshold, not a measured one, so every design's longest run is released. Ten is the strict end of a figure its source gives as a possible explanation of its own observation — "this could be because RNase H1 requires a minimum length of 7 to 10 RNA:DNA hybridized nucleotides to bind with its hybrid binding domain" — rather than as a measured minimum.[45] The qualifier does not transfer cleanly: the run counted here is the whole contiguous duplex, of which exactly six nucleotides are the RNA:DNA pairs of the gap and the rest are LNA:RNA wing pairs the source's wording would not count. Ten here is therefore a total-duplex length, while the source's 7 to 10 is a count of RNA:DNA nucleotides that this architecture holds at six in every design, below the range the source states; what relation the one criterion bears to the other is not established here. The count it produces is a floor at that choice, and what the choice costs is stated beside the count in §2.5.

The null for this screen. Bounded by six transcripts, this screen has no chance expectation of its own, and a count against six transcripts is not interpretable without one. Six ensembles of arbitrary 16-mers were therefore pushed through the identical screen, 200 draws per design in each: each design's target window shuffled to preserve base composition; shuffled to preserve dinucleotide composition by an Eulerian-path shuffle; drawn from uniform bases; drawn from the panel's pooled base composition; and two arms holding either the catalytic gap or the wings fixed while shuffling the other. A seventh is structural rather than a chance null: a random window of a real donor parent joined to a random window of real *NR4A3* at one of the junction offsets the panel's own registers use, which keeps the whole design rule and randomises only where in each transcript the two pieces come from. Proportions carry Wilson 95% intervals. The pseudo-random stream is written out in the released code rather than taken from the interpreter's own, so the artefact is reproducible bit for bit. §2.5 reports the result.

5. **The genome scan.** Screens 1 to 4 are bounded either by an annotation or by six transcripts. The fifth removes that bound. Every distinct target window and its reverse complement was tested against every position of GRCh38 in both orientations at ≤2 mismatches, exhaustively: 2,948,609,696 windows over a measured 3.10×10^9 nucleotides, with no seed, no word size and therefore no search sensitivity to quantify. §2.7 reports it.

Strand orientation. A match matters only if an antisense oligonucleotide could base-pair with it, which means the sense strand; a window carrying the reverse complement is not a liability at all. `blastn` searches both strands, so such a hit passes an identity filter unless orientation is parsed, and screens produced before that parsing was added recorded them as cleavage risks. Orientation is now parsed and filtered in all 38 junction screens and the 183 designs they hold, and therefore in every cleanliness statement made here. Only two released screens are unfiltered, and neither carries a junction or supports a claim here (SI §S5). The same rule governs pre-mRNA, which is transcribed in transcript orientation: a forward match can be base-paired and a reverse-complement match cannot.

A design is called clean where it carries no sense-strand near-match, and that is always a statement about a complete hit list at a stated search depth. Both qualifications matter. A hit list the cap truncated is not complete, so no verdict is available for that design, and a design clean at one depth need not be clean at another. §2.4 and §5 report both effects.

Target-site accessibility. Estimated as mean unpaired probability over a local fold of up to 180 nucleotides, computed with the ViennaRNA partition function,[46] and spanning 0.160 to 0.707 across all 190 designs at real exon junctions with a median of 0.477. It is released with the artefacts and ranks nothing here. That omission is deliberate — accessibility bears on potency, which is not claimed for any sequence, rather than on the discrimination this work is about — and SI §S1 gives the three reasons in full.

Expression of the off-target loci. No screen above says whether a matched gene is transcribed where the drug goes. For four of the five junctions with a published exon-resolved breakpoint — those Table 6 covers — the gene loci their deeper screens return in the gap-paired class were read against GTEx v8 median TPM.[47] No such reading was taken at *TFG* exon 7, so that junction carries no expression reading rather than a negative one. The readings are in two blocks, reported separately and never combined, against two cuts used for legibility rather than as thresholds of concern: a lower cut of 1 TPM, below which a reading is taken as below detection, and an upper cut of 10 TPM. The first block is liver and both kidney compartments, the organs a systemically dosed phosphorothioate gapmer distributes to; the second is six soft-tissue types, standing in for the compartment EMC arises in, since no atlas contains the tumour itself. NCBI Gene supplied locus identity, so a locus with no reading is attributed rather than left blank, and the Human Protein Atlas[48] was read as a transport check only, its consensus incorporating GTEx rather than confirming it independently.

Discrimination model. The binary assumption that any mismatch inside the gap abolishes cleavage is not supported by the primary literature and is not used for any claim of cleanliness. The field's general figure for single-nucleotide discrimination by a gapmer carrying no positional modification in its gap is approximately five-fold,[33] and at 16-mer length one study reports no efficient discrimination at all.[49] Both are measured against a single-nucleotide substitution rather than a fusion junction, and the pessimistic one used unmodified antisense DNA, so they are used here as bounds for unmodified chemistry rather than as a property of this architecture. Gapmer-specific work points the same way, which is why the bounds are not narrowed: across more than 120 gapmers spanning five single-nucleotide changes, only two or three achieved preferential cleavage of the mutant allele in cells,[50] and where allele selectivity is achieved it is engineered by modifying a gap position to block cleavage of the near-match rather than obtained from the mismatch itself.[51]

Every screen that resolves the gap was therefore re-scored under both bounds as a graded residual cleavage load, holding the hit set fixed so that only the scoring changed: all 38 junction screens, and 39 of the 93 screens released in total (SI §S4). Two distinct bounds follow, and they run in opposite directions. Where a hit list is truncated, the strand of the remainder is unrecoverable, so some designs keep a strand-blind count; that over-counts liability, because it includes matches no antisense oligonucleotide can hybridise, and is an upper bound. The same truncation also means fewer hits are recorded than the search returned, so the count of hits is itself right-censored: a lower bound on how many exist. Each design records which bounds apply to it.

Duplex thermodynamics. A base count is a proxy for discrimination, and a free energy is the field's standard instrument for it, so each design was also scored thermodynamically. A junction gapmer is a perfect complement of the fusion across all 16 positions, while a parent transcript can pair only the half of the oligonucleotide it contributes. The comparison is therefore the full 16-mer duplex against the donor-side and acceptor-side runs alone. Nearest-neighbour enthalpies and entropies for a DNA:RNA hybrid were taken from Sugimoto and colleagues,[52] and $\Delta G°37$ computed as $\Delta H° - T\Delta S°$. The arithmetic was checked against an independent implementation, which agreed exactly; that check verifies the summation and not the choice of strand, and the strand concentration it uses enters no reported free energy (SI §S2).

These designs carry LNA wings and the table is for an unmodified hybrid, so what is computed is the duplex the DNA backbone would form. Because the junction lies inside the gap, each parent pairs one of the two five-nucleotide LNA wings while the fusion pairs both, so LNA should widen this margin rather than narrow it and every reported value is a conservative floor. That direction follows from the architecture and was not computed: no LNA parameters were applied.

Conventional design rules. Every design was separately audited against four conventional antisense design rules: GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, and no CpG dinucleotide. The audit is not there to grade the designs, but to ask whether conventional triage and the gap-level margin would select the same molecules.

Availability. All code, graded artefacts and per-design tables are released under a single archived version, deposited from the public repository at `github.com/trimcrae/Rare-cancers` [ARCHIVE DOI]. Every result reported here is re-derived from the committed artefacts in that archive without network access or credentials. Regenerating the specificity screens from scratch is not offline, because the alignment screen queries NCBI BLAST and the exhaustive transcript scan downloads the GRCh38.p14 RefSeq RNA set, but no reported number requires it: each screen's hit set is archived and the re-scores hold it fixed. The pre-mRNA and mature-parent screens are fully offline against the archive, since the retrieved unspliced sequence and exon coordinates travel with it.

Tables

Table 1. The in-frame junction space across five *NR4A3* fusion partners. Every donor-exon × *NR4A3*-acceptor-exon pair was graded against the frame condition before any design was emitted. The gap-level margin is the number of junction-unique bases inside the six-nucleotide catalytic gap on the shorter side of the junction. Frame compatibility is an arithmetic property of exon structure and is not a claim about which junctions patients carry.

5′ partner	donor exons	exon pairs graded	in-frame	with ≥1 fusion-specific design	GC range of those designs (%)	best gap-level margin
EWSR1	17	51	8	8	37.5–75.0	3
FUS	15	45	8	8	31.2–62.5	3
TAF15	16	48	8	8	31.2–68.8	3
TCF12	21	63	8	8	25.0–56.2	3
TFG	8	24	6	6	25.0–50.0	3
all 5 partners	—	**231**	**38**	**38**	—	—

Table 2. Predicted specificity per screened junction. One row per junction; figures are for the design with the highest gap-level margin at that junction, which is the ranking the Methods define, and NOT for that junction's cleanest design — the two are often different molecules, and the cleanest ones are in Table 3. Near-match counts are of RefSeq transcript accessions and are also given collapsed to distinct gene loci, since RefSeq carries one accession per annotated variant. A "≥" marks a right-censored count: the screens store the top 15 hits per design, so a design with more is a lower bound. All 38 junction screens are filtered by alignment orientation. `XM_`/`XR_` records are computationally predicted gene models rather than curated transcripts, and are counted separately for that reason. None of these numbers is a measurement of off-target activity.

[1] A near-match count is what the search returned on EITHER strand; a match on the strand opposite the target window cannot be hybridised by an antisense oligonucleotide and is not a liability. Across this corpus 44% of apparent gap-spanning hits (738 of 1,677) are of that kind, which is why the two columns differ and why the raw count alone should not be read as load. This column counts only the 15 RETAINED hits. The gap-spanning locus column is recounted from those hits wherever they are the complete list, and is exact there; a "≤" marks a truncated design, where the column instead carries the screen's own count over every ranked hit, computed under a locus assignment since corrected that split some genes across accessions and therefore over-counts. The two columns are not in conflict where a truncated design shows "≥0" sense-strand hits and a non-zero gap-spanning locus count: the sense-strand hits are real and simply fall outside the stored window, which is precisely why such a design cannot be called clean.

[2] Counted over the gap-spanning loci only, not over all of that design's near-match loci.

[3] The same design re-screened at a tenfold deeper alignment ceiling, with retention raised to match it so that no hit list is truncated. Because no list is truncated, the gap-spanning locus column at this depth is recounted from the complete stored hits under the current locus assignment and is exact; it is not the screen's own stored figure, which was computed before that assignment was corrected and splits any gene whose description carries a comma across one accession per transcript variant. It is therefore the same quantity, counted the same way, as the locus figures in Table 4 and in the Results. The three columns are the counterparts of the default-depth columns to their left, given beside them rather than in place

of them because the default depth is where the corpus-wide counts elsewhere in the paper were computed and the two must stay comparable. Read together they are the paper's censoring result at the level of a single row: a default-depth count is a lower bound whether or not it reached the 50-hit cap, and three junctions whose default cell reads zero in the gap-spanning column carry gap-spanning hits at ten times the depth. A “—” means the deeper re-screen returned no result for that design and is not a count of zero; three of the panel's 190 records failed at this ceiling.

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand¹	loci with a gap-spanning hit	of those, predicted models only²	at the deeper ceiling: near-matches³	of those, on the sense strand³	loci with a gap-spanning hit³	≤1-mismatch matches across that junction's designs, median (max)
EWSR1 e10::NR4A3 e3	5	3	5′-GGGCATATCTAGATCA-3′	35 → ≥2	≥15	≤4	0	138	137	5	4 (32)
EWSR1 e12::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
EWSR1 e13::NR4A3 e3	5	3	5′-GGGCATATCTCCACGG-3′	31 → ≥3	≥12	≤2	0	63	47	2	2 (25)
EWSR1 e15::NR4A3 e3	5	3	5′-GGGCATATCCGGGGGC-3′	36 → ≥2	≥12	≤1	0	93	62	1	1 (10)
EWSR1 e1::NR4A3 e3	5	3	5′-GGGCATATCCGTGGAC-3′	0 → 0	0	0	0	27	18	0	0 (0)
EWSR1 e4::NR4A3 e3	5	3	5′-GGGCATATCAGTGGGA-3′	11 → 4	11	3	2	90	67	8	4 (10)
EWSR1 e7::NR4A3 e3	5	3	5′-GGGCATATTCTGCTGC-3′	32 → ≥9	≥6	≤2	0	300	65	6	9 (21)
EWSR1 e9::NR4A3 e3	2	3	5′-GGGCATATCACCAGGC-3′	29 → ≥2	≥6	≤2	0	165	81	7	1 (22)
FUS e10::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
FUS e11::NR4A3 e3	5	3	5′-GGGCATATCTCCTCGC-3′	30 → ≥1	≥15	≤1	0	60	30	1	0 (3)
FUS e13::NR4A3 e3	5	3	5′-GGGCATATCCATGTGA-3′	6 → 5	2	2	1	20	8	2	1 (9)
FUS e1::NR4A3 e3	5	3	5′-GGGCATATCGTTTGAG-3′	≥50 → ≥7	≥13	≤1	0	129	116	2	10 (38)

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand¹	loci with a gap-spanning hit	of those, predicted models only²	at the deeper ceiling: near-matches³	of those, on the sense strand³	loci with a gap-spanning hit³	≤1-mismatch matches across that junction's designs, median (max)
FUS e3::NR4A3 e3	4	3	5′-GGGCATATTGTTCTGG-3′	18 → ≥4	≥1	≤2	0	148	34	4	3 (23)
FUS e5::NR4A3 e3	5	3	5′-GGGCATATCTCCACCT-3′	32 → ≥8	≥8	≤2	1	127	47	4	2 (3)
FUS e7::NR4A3 e3	5	3	5′-GGGCATATCACCAAAT-3′	34 → ≥5	≥8	≤0	0	141	107	4	12 (39)
FUS e8::NR4A3 e3	5	3	5′-GGGCATATCGGAGTCA-3′	4 → 3	3	1	0	4	3	1	0 (1)
TAF15 e11::NR4A3 e3	5	3	5′-GGGCATATCATCAAAC-3′	9 → 4	6	1	1	189	141	6	2 (22)
TAF15 e12::NR4A3 e3	4	2	5′-AGGGCATATCTCGCCG-3′	42 → ≥4	≥11	≤3	0	42	33	3	3 (4)
TAF15 e14::NR4A3 e3	5	3	5′-GGGCATATCTCCTCCT-3′	≥50 → ≥2	≥11	≤2	0	174	95	6	19 (40)
TAF15 e1::NR4A3 e3	5	3	5′-GGGCATATCCGACATG-3′	5 → 2	0	0	0	5	0	0	0 (0)
TAF15 e4::NR4A3 e3	5	3	5′-GGGCATATCTGACTGA-3′	34 → ≥3	≥15	≤4	1	78	57	7	10 (53)
TAF15 e6::NR4A3 e3	5	3	5′-GGGCATATCTTGTGTG-3′	11 → 7	7	4	2	62	10	5	2 (2)
TAF15 e8::NR4A3 e3	5	3	5′-GGGCATATCACCAAAA-3′	41 → ≥4	≥13	≤2	0	133	93	3	10 (17)
TAF15 e9::NR4A3 e3	5	3	5′-GGGCATATCAGCATCT-3′	23 → ≥7	≥0	≤2	0	68	48	5	1 (4)
TCF12 e11::NR4A3 e3	5	3	5′-GGGCATATCTAGAATG-3′	15 → 5	2	1	1	46	17	4	2 (6)
TCF12 e13::NR4A3 e3	5	3	5′-GGGCATATCTGTGAGA-3′	17 → ≥8	≥6	≤1	1	170	80	4	1 (12)
TCF12 e17::NR4A3 e3	5	3	5′-GGGCATATCTCTATAA-3′	8 → 3	0	0	0	118	101	5	1 (6)
TCF12 e19::NR4A3 e3	5	3	5′-GGGCATATCTCTGACT-3′	12 → 4	8	1	1	102	79	3	1 (2)

junction	designs screened	best gap-level margin	that design	near-matches, either strand (transcripts → loci)	of the retained hits, on the sense strand[1]	loci with a gap-spanning hit	of those, predicted models only[2]	at the deeper ceiling: near-matches[3]	of those, on the sense strand[3]	loci with a gap-spanning hit[3]	≤1-mismatch matches across that junction's designs, median (max)
TCF12 e3::NR4A3 e3	5	3	5′-GGGCATATCTGATCCA-3′	15 → 4	10	2	2	374	246	6	2 (35)
TCF12 e5::NR4A3 e3	5	3	5′-GGGCATATCCATCAGA-3′	26 → ≥1	≥15	≤17	0	83	70	1	2 (32)
TCF12 e7::NR4A3 e3	4	2	5′-GGCATATCAAGCGCTG-3′	2 → 2	0	0	0	2	0	0	0 (1)
TCF12 e9::NR4A3 e3	4	3	5′-GGGCATATCTTGCATA-3′	14 → 2	8	1	0	86	64	6	8 (23)
TFG e2::NR4A3 e3	5	3	5′-GGGCATATCTTCATCT-3′	≥50 → ≥1	≥15	≤2	0	207	117	5	35 (87)
TFG e3::NR4A3 e3	5	3	5′-GGGCATATCAAATAAT-3′	14 → 6	9	3	0	217	161	6	9 (17)
TFG e4::NR4A3 e3	5	3	5′-GGGCATATCATTTTCA-3′	≥50 → ≥3	≥15	≤5	2	318	238	15	41 (100)
TFG e5::NR4A3 e3	5	3	5′-GGGCATATCTGAAACC-3′	41 → ≥6	≥11	≤7	1	112	72	10	3 (21)
TFG e6::NR4A3 e3	5	3	5′-GGGCATATCTTCAATC-3′	37 → ≥3	≥2	≤0	0	238	193	3	11 (21)
TFG e7::NR4A3 e3	5	3	5′-GGGCATATCTGAATAC-3′	27 → ≥3	≥3	≤3	1	43	12	4	4 (26)

Table 3. The 9 designs with no sense-strand near-match at the default search depth. Six of these lose the property when the same junctions are re-screened at a tenfold deeper alignment ceiling, three of them having returned no near-match at all here; §2.4 reports that measurement and names the three that survive it. This table is the default-depth result, retained because it is the depth at which the corpus-wide counts elsewhere in the paper were computed. Every design at the 6 junctions where one exists. A design qualifies only if its retained hit list is not truncated — no more near-matches than the 15 the screens store — because the strand of an unstored hit cannot be recovered, so a truncated list cannot establish that nothing on the sense strand remains. The underlying search is itself capped, so these are the designs whose near-match lists are shortest, not the designs whose lists are known to be exhaustive. ΔΔG°37 is the margin by which the fusion duplex is favoured over the best duplex either parent can form, for an unmodified DNA:RNA hybrid; because the fusion duplex pairs both LNA wings and each parent duplex only one, it is a lower bound on the modified oligonucleotide's discrimination rather than an upper one. None of these numbers is a measurement of off-target activity, and none speaks to cleavage.

[4] Under the optimistic five-fold and the pessimistic no-discrimination bound on RNase-H1 single-mismatch discrimination. A single value means the two bounds agree.

[5] Of four conventional antisense design rules: GC within 40–60%, no G-quadruplex motif, no homopolymer run of four, no CpG dinucleotide.

[6] Whether the design still carries no sense-strand near-match once its junction is re-screened at the tenfold deeper ceiling. The verdict is computed from the three deep columns beside it, not asserted, so this table cannot come to disagree with §2.4 about which designs survive. The six that do not are the reason this table's default-depth zeros must not be read on their own.

design	junction	GC (%)	gap-level margin	ΔΔG°37 (kcal/mol)	near-matches, either strand	of those, on the sense strand	exact / ≤1-mismatch matches	residual cleavage load, both bounds[4]	conventional rules failed[5]	at the deeper ceiling: near-matches	of those, on the sense strand	loci with a gap-spanning hit	survives[6]
5′-GCATATCCGTGGACGC-3′	EWSR1 e1::NR4A3 e3	62.5	1	7.981	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	84	83	1	**no**
5′-GGCATATCCGTGGACG-3′	EWSR1 e1::NR4A3 e3	62.5	2	10.085	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	29	22	0	**no**
5′-GGGCATATCCGTGGAC-3′	EWSR1 e1::NR4A3 e3	62.5	3	12.189	0	0	0 / 0	0	GC outside 40–60%, contains a CpG	27	18	0	**no**
5′-AGGGCATATCGGAGTC-3′	FUS e8::NR4A3 e3	56.2	2	10.895	3	0	0 / 0	0	contains a CpG	3	0	0	yes
5′-GGGCATATCCGACATG-3′	TAF15 e1::NR4A3 e3	56.2	3	11.894	5	0	0 / 0	0	contains a CpG	5	0	0	yes
5′-GGGCATATCTCTATAA-3′	TCF12 e17::NR4A3 e3	37.5	3	8.556	8	0	0 / 0	0	GC outside 40–60%	118	101	5	**no**
5′-GCATATCAAGCGCTGC-3′	TCF12 e7::NR4A3 e3	56.2	1	7.98	1	0	0 / 0	0	contains a CpG	18	2	0	**no**
5′-GGCATATCAAGCGCTG-3′	TCF12 e7::NR4A3 e3	56.2	2	10.085	2	0	0 / 0	0	contains a CpG	2	0	0	yes
5′-CAGGGCATATCTTGCA-3′	TCF12 e9::NR4A3 e3	50.0	1	9.325	7	0	0 / 0	0	none	67	18	4	**no**

Table 4. The best available design at each of the 38 in-frame junctions. Tables 2 and 3 select across the panel; this table selects within each junction, which is the question a patient's fusion poses. Designs are ranked by parent liability first, since sparing the wild-type parents is what the modality exists for, then by pre-mRNA sites, then by distinct gene loci, with ties broken on gap-level margin rather than on raw hit counts. Nothing was re-screened: every field is joined from a screen already reported above. Whether a junction has a published exon-resolved breakpoint is reported separately from specificity and never folded into the ranking — “published” means an exon-resolved EMC breakpoint is reported for that exon pair, “exon not reported” that the partner has a resolved breakpoint at a different exon, and “none published” that no exon-resolved breakpoint has been reported for that partner at all. The last of those is absence of evidence: EMC case reports usually name the partner gene without sequencing to nucleotide resolution. Gap-paired near-matches are at the tenfold deeper alignment ceiling, where every hit list is complete. The genome column is the observed number of gap-paired sites at ≤2 mismatches over the number expected for an arbitrary 16-mer, so 1.00 is chance. A junction with no design clearing the parent screen is reported as such rather than given a best row.

junction	exon-resolved breakpoint	designs clearing the parent screen	best available design	gap-level margin	longest parent duplex through the gap (bp)	gap-paired near-matches at the deeper ceiling (transcripts → loci)	genome-wide gap-paired load, observed/expected
EWSR1 e10::NR4A3 e3	exon not reported	3 of 5	5′-GGCATATCTAGATCAA-3′	2	7	70 → 4	0.61
EWSR1 e12::NR4A3 e3	published	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
EWSR1 e13::NR4A3 e3	published	3 of 5	5′-GGGCATATCTCCACGG-3′	3	8	24 → 2	0.29
EWSR1 e15::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCCGGGGGC-3′	3	0	33 → 1	0.06
EWSR1 e1::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCCGTGGAC-3′	3	0	0 → 0	0.07
EWSR1 e4::NR4A3 e3	exon not reported	4 of 5	5′-AGGGCATATCAGTGGG-3′	2	6	9 → 3	0.71
EWSR1 e7::NR4A3 e3	exon not reported	5 of 5	5′-CAGGGCATATTCTGCT-3′	1	8	14 → 5	1.08
EWSR1 e9::NR4A3 e3	exon not reported	4 of 5	5′-GGCATATCACCAGGCT-3′	2	7	15 → 4	0.73
FUS e10::NR4A3 e3	none published	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
FUS e11::NR4A3 e3	none published	3 of 5	5′-GGGCATATCTCCTCGC-3′	3	9	30 → 1	0.22
FUS e13::NR4A3 e3	none published	4 of 5	5′-GGGCATATCCATGTGA-3′	3	7	2 → 2	0.75
FUS e1::NR4A3 e3	none published	2 of 5	5′-GGGCATATCGTTTGAG-3′	3	8	5 → 2	0.12
FUS e3::NR4A3 e3	none published	5 of 5	5′-GGCATATTGTTCTGGC-3′	2	7	18 → 3	0.76
FUS e5::NR4A3 e3	none published	2 of 3	5′-GGGCATATCTCCACCT-3′	3	8	33 → 4	0.65
FUS e7::NR4A3 e3	none published	4 of 5	5′-GGCATATCACCAAATT-3′	2	7	10 → 3	0.92
FUS e8::NR4A3 e3	none published	3 of 5	5′-AGGGCATATCGGAGTC-3′	2	0	0 → 0	0.06
TAF15 e11::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCATCAAAC-3′	3	8	123 → 6	0.62
TAF15 e12::NR4A3 e3	exon not reported	3 of 5	5′-GGGCATATCTCGCCGC-3′	3	6	4 → 2	0.04
TAF15 e14::NR4A3 e3	exon not reported	0 of 5	—	—	—	—	—

junction	exon-resolved breakpoint	designs clearing the parent screen	best available design	gap-level margin	longest parent duplex through the gap (bp)	gap-paired near-matches at the deeper ceiling (transcripts → loci)	genome-wide gap-paired load, observed/expected
TAF15 e1::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCCGACATG-3′	3	0	0 → 0	0.04
TAF15 e4::NR4A3 e3	exon not reported	2 of 5	5′-GGCATATCTGACTGAC-3′	2	8	12 → 2	0.69
TAF15 e6::NR4A3 e3	published	1 of 5	5′-GGGCATATCTTGTGTG-3′	3	9	8 → 5	0.60
TAF15 e8::NR4A3 e3	exon not reported	4 of 5	5′-GGGCATATCACCAAAA-3′	3	7	36 → 3	0.86
TAF15 e9::NR4A3 e3	exon not reported	2 of 5	5′-AGGGCATATCAGCATC-3′	2	6	6 → 2	1.13
TCF12 e11::NR4A3 e3	exon not reported	3 of 5	5′-GGCATATCTAGAATGC-3′	2	8	4 → 3	0.69
TCF12 e13::NR4A3 e3	exon not reported	2 of 5	5′-GGCATATCTGTGAGAG-3′	2	7	6 → 5	1.08
TCF12 e17::NR4A3 e3	exon not reported	3 of 5	5′-GGGCATATCTCTATAA-3′	3	7	14 → 5	0.83
TCF12 e19::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCTCTGACT-3′	3	8	75 → 3	0.82
TCF12 e3::NR4A3 e3	exon not reported	0 of 5	—	—	—	—	—
TCF12 e5::NR4A3 e3	published	4 of 5	5′-GGGCATATCCATCAGA-3′	3	7	17 → 1	0.56
TCF12 e7::NR4A3 e3	exon not reported	2 of 5	5′-GGGCATATCAAGCGCT-3′	3	8	0 → 0	0.26
TCF12 e9::NR4A3 e3	exon not reported	1 of 5	5′-GGGCATATCTTGCATA-3′	3	8	39 → 6	0.71
TFG e2::NR4A3 e3	none published	0 of 4	—	—	—	—	—
TFG e3::NR4A3 e3	none published	4 of 5	5′-GGGCATATCAAATAAT-3′	3	8	18 → 6	1.35
TFG e4::NR4A3 e3	none published	1 of 5	5′-AGGGCATATCATTTTC-3′	2	7	82 → 12	1.77
TFG e5::NR4A3 e3	none published	2 of 5	5′-GGCATATCTGAAACCT-3′	2	8	34 → 7	1.38
TFG e6::NR4A3 e3	none published	3 of 5	5′-GGGCATATCTTCAATC-3′	3	9	29 → 3	0.73
TFG e7::NR4A3 e3	published	2 of 5	5′-GGCATATCTGAATACT-3′	2	9	24 → 6	1.24

Table 5. Gap length against junction specificity, at one junction and across the design space. The same junctions tiled and screened at three gapmer geometries, wing held at five nucleotides so that only the catalytic gap changes. Inside the gap the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side are complements: they sum to the gap, which the generating module asserts for every design rather than assuming. Within one geometry the gap is fixed, so each nucleotide of gap-level margin is one FEWER nucleotide of contiguous wild-type-parent duplex; what raises the parent-paired run at every register is a longer gap, which is the only way past a geometry's ceiling of half its gap rounded down. The two directions are reported separately and never combined into a score. Near-match counts fall partly for a reason the instrument guarantees rather than measures: at a fixed budget of two mismatches every locus a longer design can reach is also reached by each of its own shorter sub-windows, so the set can only shrink, and two mismatches is a fractionally stricter test at 20 nucleotides than at 16. Only the size of the fall and which designs reach zero are measurements. The three blocks carry different denominators and are not comparable across blocks: the junction block is one molecule, the matched-junction block is the six junctions every geometry was screened at, and the corpus block is each geometry's whole design space, which is not screened at the same junctions. The exhaustive GRCh38 genome scan is unavailable at 18 and 20 nucleotides by construction, so no row reports it. Two of the corpus rows carry a ten-base-pair criterion and they are not the same measurement. "…and that duplex reaches ten base pairs" is the mature-parent screen, a search over every window of all six parent transcripts, and it is the row §2.5's 87 of 190 and §2.9's 87 / 88 / 87 are read from. "At the design's own seam" is arithmetic on the junction itself: because the wing is five throughout, a parent's hybrid at that seam is five base pairs plus its share of the gap, so pairing five nucleotides of contiguous gap DNA and reaching a ten-base-pair seam hybrid are the same condition and are reported as one row. ΔG°37 values are for an unmodified DNA:RNA hybrid; the wing is five at every geometry, so LNA affinity enters each parent duplex identically and cannot explain a difference between the columns. None of these numbers is a measurement of cleavage.

	5-6-5 (16-mer)	5-8-5 (18-mer)	5-10-5 (20-mer)
At the *EWSR1* e12 / *TAF15* e11 / *FUS* e10 junction			
design (5′→3′)	GGGCATATCATCAAAC	AGGGCATATCATCAAACC	CAGGGCATATCATCAAACCA
gap-level margin	3	4	5
sense-strand gap-spanning cleavage risks	123	3	0
gene loci carrying one	6	1	0
near-matches (≤2 mismatches, deeper ceiling)	189	50	20
≤1-mismatch matches over 186,185 transcripts	1	0	0
mature-parent duplex through the whole gap (bp)	8 (*TFG*)	0	0
contiguous DNA a wild-type parent pairs (nt)	3	4	5
most stable parent ΔG°37 (kcal/mol)	−7.77	−8.66	−10.25
Over the six junctions screened at every geometry			
designs screened	30	42	54
median near-matches	86.5	15	0
median gap-spanning cleavage risks	21	0	0
designs carrying none	8 of 30	28 of 42	54 of 54
most risk loci on any one design	7	2	0
designs with no near-match at all	0 of 30	7 of 42	39 of 54
Over each geometry's whole design space			
junction-spanning registers per junction	5	7	9
fusion-specific designs	190	266	342

	5-6-5 (16-mer)	5-8-5 (18-mer)	5-10-5 (20-mer)
best gap-level margin available	3	4	5
a mature parent can pair the whole gap	181 of 190	130 of 266	87 of 342
…and that duplex reaches ten base pairs, the criterion applied throughout	87 of 190	88 of 266	87 of 342
at the design's own seam, the parent pairs ≥5 nt of contiguous gap DNA	76 of 190	228 of 266	342 of 342
designs pairing the gap in parent pre-mRNA	19 of 190	11 of 266	9 of 342
median most stable parent ΔG°37 (kcal/mol)	−8.66	−11.60	−14.58

Table 6. Where the clinically-relevant reagents' off-target loci are expressed. Every gene locus returned by the deeper screens at four of the five junctions with a published exon-resolved EMC breakpoint, read against reference expression data. The two compartments answer different questions and are never combined: a systemically dosed phosphorothioate gapmer distributes predominantly to liver and kidney, so liver, kidney - cortex and kidney - medulla address exposure, while the soft-tissue column is the normal tissue of the compartment EMC arises in and stands in for a tumour no reference atlas contains. Values are GTEx v8 median TPM across each tissue's donors. The two cuts behind the last column are stated for legibility and are not thresholds of concern: below 1 TPM in all three exposure tissues reads as below detection, at or above 10 TPM in any of them as the level at which an off-target hypothesis would have to be tested. Every raw median is released so another cut can be applied without re-running. Tiling registers is how many of the designs tiled across that junction return the locus, which is robustness to where the window is placed and is a different axis from the record count beside it; neither is ranked on. Transcript records are how many accessions RefSeq lists for the gene, that is annotation depth, not expression and not affinity. A locus with no reading carries the reason rather than a zero, because an absent reading is not a reading of absence. Every hit behind this table sits at 14 of 16 identity, the loosest the screen admits, so nothing here distinguishes these loci from one another on affinity. None of these numbers is a measurement of cleavage, and no expression figure is a predicted cleavage event.

junction	gene locus	transcript records	tiling registers returning it	Liver	Kidney - Cortex	Kidney - Medulla	soft-tissue proxy maximum	exposure-organ reading
EWSR1 e12::NR4A3 e3	*ANKS1B*	67	1 of 1	0.03	0.46	0.28	3.6 (Artery - Tibial)	below the lower cut in all three
	ZNF667	37	1 of 1	0.31	1.63	2.58	6.2 (Nerve - Tibial)	detectable, below the upper cut
	GMCL1	9	1 of 1	4.52	4.98	6.72	18.3 (Artery - Tibial)	detectable, below the upper cut
	LOC105374140	5	1 of 1	—	—	—	—	no gene model — not measurable
	LOC105370997	4	1 of 1	—	—	—	—	no gene model — not measurable
	CHST5	1	1 of 1	0.07	0.35	0.78	0.8 (Nerve - Tibial)	below the lower cut in all three
TAF15 e6::NR4A3 e3	*G3BP2*	56	2 of 5	11.92	17.03	23.12	77.0 (Cells - Cultured fibroblasts)	at or above the upper cut
	LINC02030	22	2 of 5	0.00	0.00	0.02	0.3 (Skin - Sun Exposed (Lower leg))	below the lower cut in all three
	MIR9-2HG	18	2 of 5	—	—	—	—	no reading taken
	LAMA4	13	1 of 5	1.28	3.13	6.06	268.6 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	ZFPM2	12	3 of 5	0.64	0.55	0.40	9.6 (Artery - Tibial)	below the lower cut in all three
	GNAL	10	2 of 5	0.66	1.74	1.81	8.6 (Artery - Tibial)	detectable, below the upper cut
	NRP1	5	5 of 5	6.62	16.87	17.81	104.7 (Cells - Cultured fibroblasts)	at or above the upper cut
	SLC17A3	4	2 of 5	9.14	33.61	8.09	0.0 (Adipose - Subcutaneous)	at or above the upper cut
	CA5B	3	3 of 5	0.66	1.57	1.96	11.7 (Artery - Tibial)	detectable, below the upper cut
	CA5BP1-CA5B	3	3 of 5	—	—	—	—	no reading taken

junction	gene locus	transcript records	tiling registers returning it	Liver	Kidney - Cortex	Kidney - Medulla	soft-tissue proxy maximum	exposure-organ reading
	EEFSEC	3	1 of 5	16.15	14.37	15.57	39.1 (Nerve - Tibial)	at or above the upper cut
	ANKRD26P3	1	1 of 5	0.00	0.00	0.00	0.0 (Muscle - Skeletal)	below the lower cut in all three
	GBP4	1	1 of 5	3.12	5.59	10.08	18.2 (Adipose - Subcutaneous)	at or above the upper cut
	LOC105376349	1	1 of 5	—	—	—	—	no gene model — not measurable
	LOC124907518	1	1 of 5	—	—	—	—	no gene model — not measurable
	NRXN3-AS1	1	1 of 5	—	—	—	—	no reading taken
	ST3GAL1	1	1 of 5	28.58	16.36	8.06	27.2 (Muscle - Skeletal)	at or above the upper cut
EWSR1 e13::NR4A3 e3	*FNBP1*	42	2 of 5	7.08	12.12	14.06	56.5 (Nerve - Tibial)	at or above the upper cut
	EHMT2	34	2 of 5	6.89	13.23	16.70	53.0 (Nerve - Tibial)	at or above the upper cut
	ZNF215	27	1 of 5	0.13	0.67	1.27	5.2 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	ESYT2	26	2 of 5	10.42	14.26	28.28	139.9 (Artery - Tibial)	at or above the upper cut
	LRP5L	8	2 of 5	7.83	6.32	11.16	8.0 (Nerve - Tibial)	at or above the upper cut
	CDC42SE1	6	2 of 5	22.74	44.73	78.97	187.6 (Nerve - Tibial)	at or above the upper cut
	THEMIS	6	1 of 5	0.13	0.08	0.18	0.3 (Adipose - Subcutaneous)	below the lower cut in all three
	LOC105374651	4	1 of 5	—	—	—	—	no gene model — not measurable
	ZC3H4	4	1 of 5	6.41	5.89	9.71	26.2 (Nerve - Tibial)	detectable, below the upper cut
	ERBIN	2	1 of 5	9.80	9.38	14.68	43.8 (Cells - Cultured fibroblasts)	at or above the upper cut
	ZNF236	2	1 of 5	1.68	2.29	2.90	8.7 (Nerve - Tibial)	detectable, below the upper cut
TCF12 e5::NR4A3 e3	*HNRNPA2B1*	100	2 of 5	184.12	247.24	457.30	656.6 (Nerve - Tibial)	at or above the upper cut
	PIK3CG	51	3 of 5	0.14	0.17	0.37	1.2 (Adipose - Subcutaneous)	below the lower cut in all three
	MROH2A	28	1 of 5	1.04	0.42	0.07	0.3 (Skin - Sun Exposed (Lower leg))	detectable, below the upper cut
	LINC02940	9	1 of 5	—	—	—	—	no reading taken
	EXOC2	6	1 of 5	5.27	6.54	8.87	22.5 (Nerve - Tibial)	detectable, below the upper cut
	KCNG3	4	2 of 5	0.00	0.09	0.06	1.3 (Nerve - Tibial)	below the lower cut in all three
	PLAC8	4	2 of 5	1.43	0.32	0.30	0.7 (Adipose - Subcutaneous)	detectable, below the upper cut
	EFCAB11	2	1 of 5	0.37	0.92	1.20	2.1 (Cells - Cultured fibroblasts)	detectable, below the upper cut
	LOC107984281	2	2 of 5	—	—	—	—	no gene model — not measurable
	LOC107985219	2	2 of 5	—	—	—	—	no gene model — not measurable
	LOC107987169	1	1 of 5	—	—	—	—	no gene model — not measurable
	LOC124905457	1	1 of 5	—	—	—	—	no gene model — not measurable

Table 7. Every reagent named in §4, what it costs on each screen and what it buys in coverage. The rows are in the order §4 decides them: the two lead reagents, the rungs of the coverage ladder above them, the four *NR4A3* exon-2 acceptor seams reported beside the panel, and the two contrast arms. Cumulative coverage is the coverage of the reagent set through that row, so the two leads are one rung and carry one

figure between them; it is discounted by the breakpoint distribution of a single series and is not a partner figure, and its interval is composed from each breakpoint fraction's own Wilson bound rather than from the point estimate. A bound row is what coverage would be if every remaining breakpoint of that partner were covered, which nothing measures. A row that adds nothing prints why, because the two reasons differ: the partner is absent from the 58-case cohort behind the denominator, or the partner is present and that exon pair carries no count in the measured within-partner distribution. The exon-2 acceptor rows are from the non-canonical-acceptor table and are never pooled into the panel, since the grade that excludes their junctions from the 38 is unchanged. A contrast arm carries no coverage figure and must not borrow its junction's, which is already counted a row above. Gap-paired near-matches are at the tenfold deeper alignment ceiling where every hit list is complete, and the parent duplex is the longest contiguous run containing the whole catalytic gap, at the ten-base-pair criterion applied throughout. None of these numbers is a measurement of off-target activity, and no row is a claim of efficacy.

reagent	junction	sequence	geometry	gap-level margin	gap-paired near-matches → loci at the deeper ceiling	longest mature-parent duplex through the gap	cumulative coverage	basis
lead reagent	EWSR1 e12::NR4A3 e3	5′-GGGCATATCATCAAAC-3′	5-6-5	3	123 → 6	8 bp (*TFG*)	68.4% (39.9–82.8)	single series, cumulative
lead reagent	TAF15 e6::NR4A3 e3	5′-GGGCATATCTTGTGTG-3′	5-6-5	3	8 → 5	9 bp (*TFG*)	68.4% (39.9–82.8)	single series, cumulative
coverage rung	EWSR1 e13::NR4A3 e3	5′-GGGCATATCTCCACGG-3′	5-6-5	3	24 → 2	8 bp (*TCF12*)	79.0% (50.3–89.2) (+10.6)	single series, cumulative
coverage rung	EWSR1 e7::NR4A3 e2	5′-CAGTGGGCTTCTGCTG-3′	5-6-5	2	51 → 7	8 bp (*TAF15*)	79.0% (50.3–89.2) (+0.0)	single series, cumulative
coverage bound	TCF12 e5::NR4A3 e3	5′-GGGCATATCCATCAGA-3′	5-6-5	3	17 → 1	7 bp (*EWSR1*)	98.3%	arithmetic bound
beside the panel	EWSR1 e13::NR4A3 e2	5′-AGTGGGCTCTCCACGG-3′	5-6-5	3	25 → 6	8 bp (*EWSR1*)	adds nothing	partner in the cohort, this exon pair uncounted in it
beside the panel	TAF15 e6::NR4A3 e2	5′-AGTGGGCTCTTGTGTG-3′	5-6-5	3	128 → 6	9 bp (*NR4A3*)	adds nothing	partner in the cohort, this exon pair uncounted in it
beside the panel	PGR e2::NR4A3 e2	5′-AGTGGGCTCTTCCATT-3′	5-6-5	3	51 → 14	9 bp (*NR4A3*)	adds nothing	partner absent from the cohort behind the denominator
gap-length control	EWSR1 e12::NR4A3 e3	5′-AGGGCATATCATCAAACC-3′	5-8-5	4	3 → 1	none	—	not a coverage row
margin contrast arm	EWSR1 e12::NR4A3 e3	5′-GCATATCATCAAACCA-3′	5-6-5	1	34 → 6	8 bp (*FUS*)	—	not a coverage row

Figure legends

Figures 1 to 3 are the main-text figures. Supplementary Figure S1 travels with the archive rather than with the Supporting Information, which carries no figure, and its legend is given here with theirs.

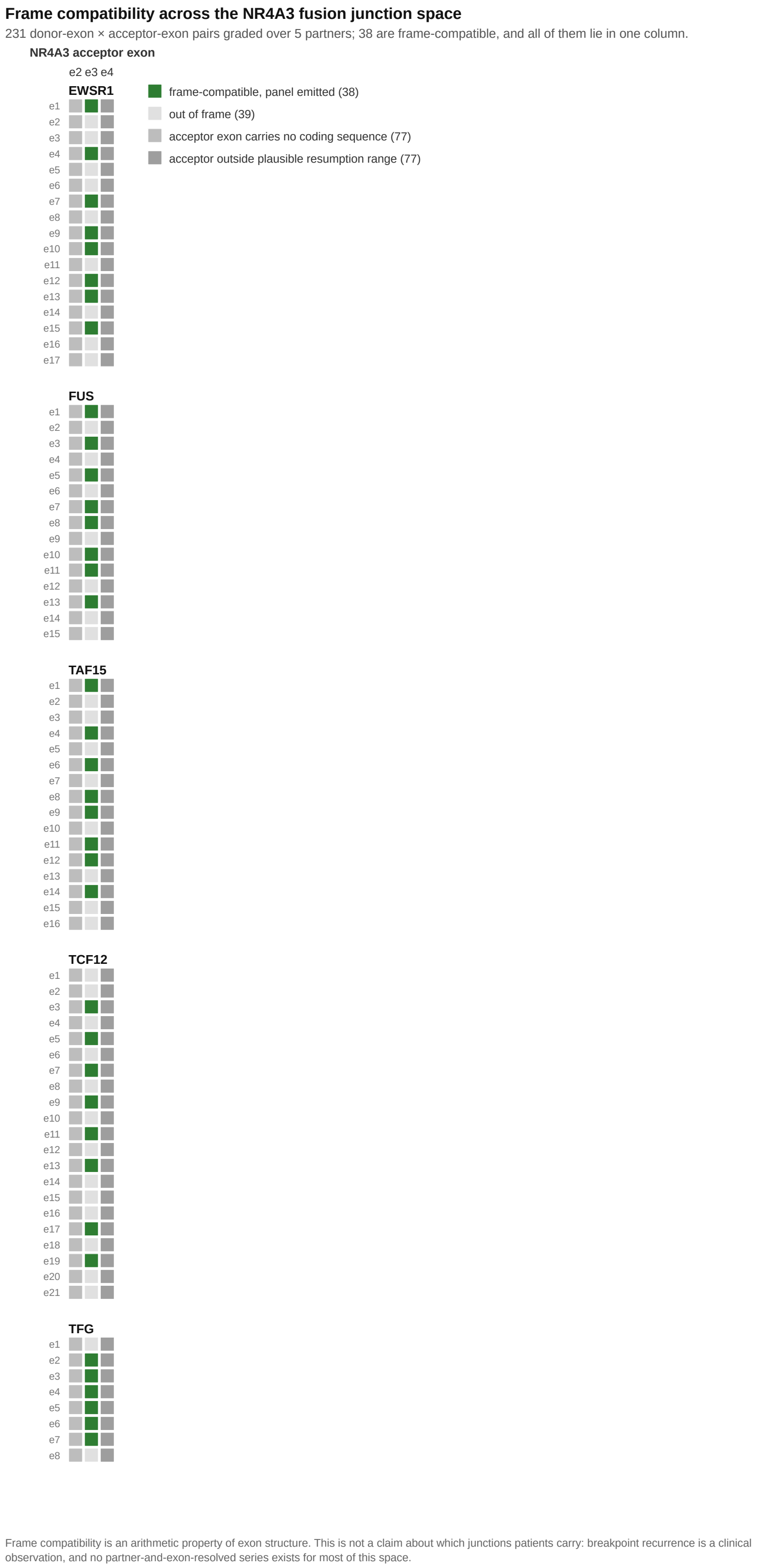


Figure 1. Reading-frame compatibility across the NR4A3 fusion junction space. All 231 donor-exon × acceptor-exon pairs across *EWSR1*, *TAF15*, *TCF12*, *FUS* and *TFG*, graded against the frame condition. Rows are donor exons grouped by partner; columns are *NR4A3* acceptor exons. Two acceptor columns are refused in every pair for structural reasons, so the 38 in-frame junctions lie in a single column.

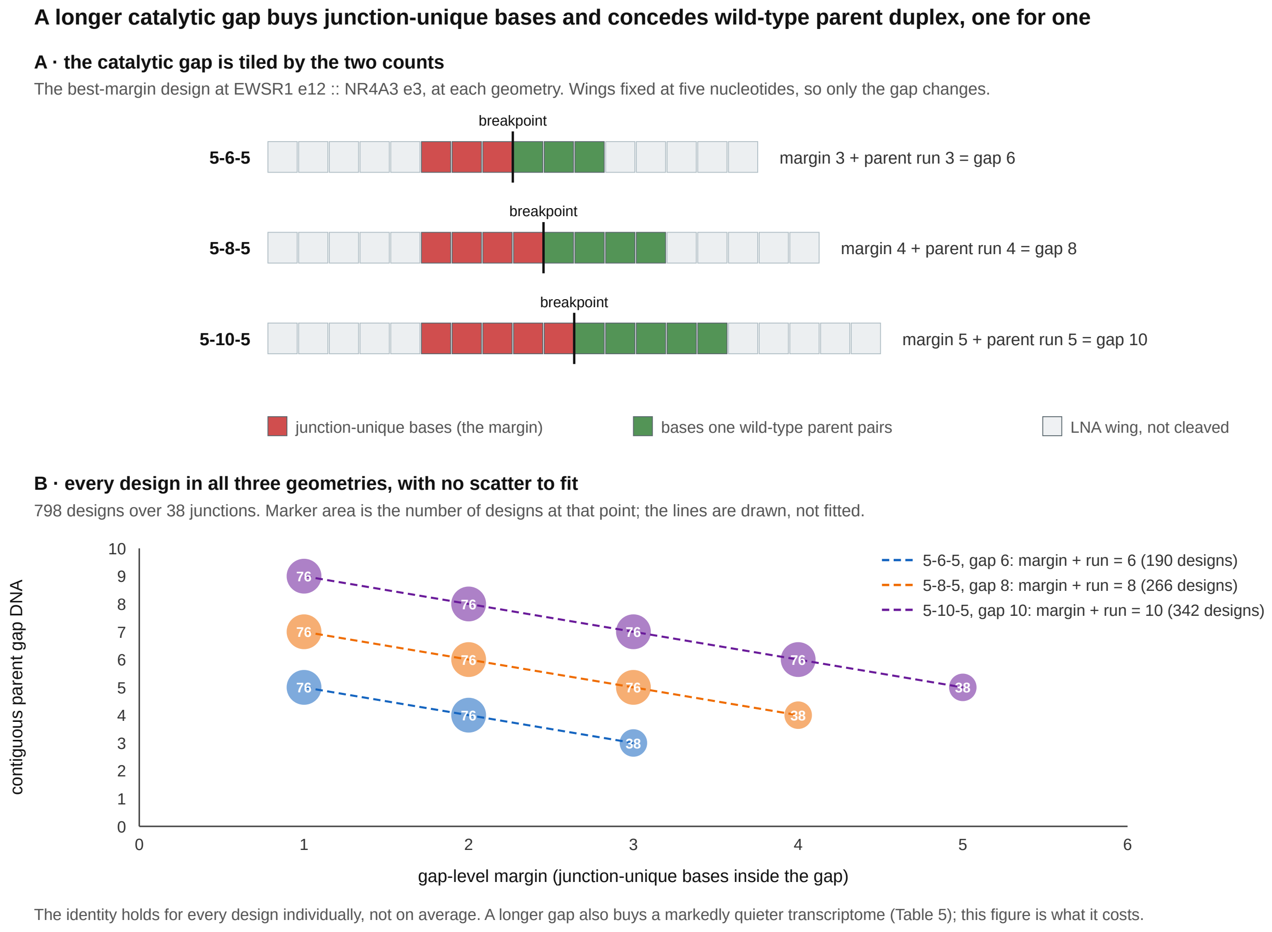


Figure 2. The margin a longer catalytic gap wins is the parent duplex it concedes. (A) The best-margin design at *EWSR1* exon 12 joined to *NR4A3* exon 3, drawn at 5-6-5, 5-8-5 and 5-10-5 with the wings held at five nucleotides. Every base inside the catalytic gap comes from the donor exon or from the acceptor exon, so the junction-unique bases on the shorter side and the bases one wild-type parent pairs on the longer side tile the gap and sum to it. (B) Every fusion-specific design in all three geometries, 798 over 38 junctions, plotted as gap-level margin against the contiguous run of gap DNA a wild-type parent can pair. Marker area is the number of designs at that point and the label is that count; the three lines are drawn from the identity, not fitted, and it holds for each design individually rather than on average. Within one geometry the two move inversely along a line of slope −1; a geometry's ceiling on margin is half its gap rounded down, and clearing it means a longer gap and a higher parent-paired run at every register (§2.9, Table 5).

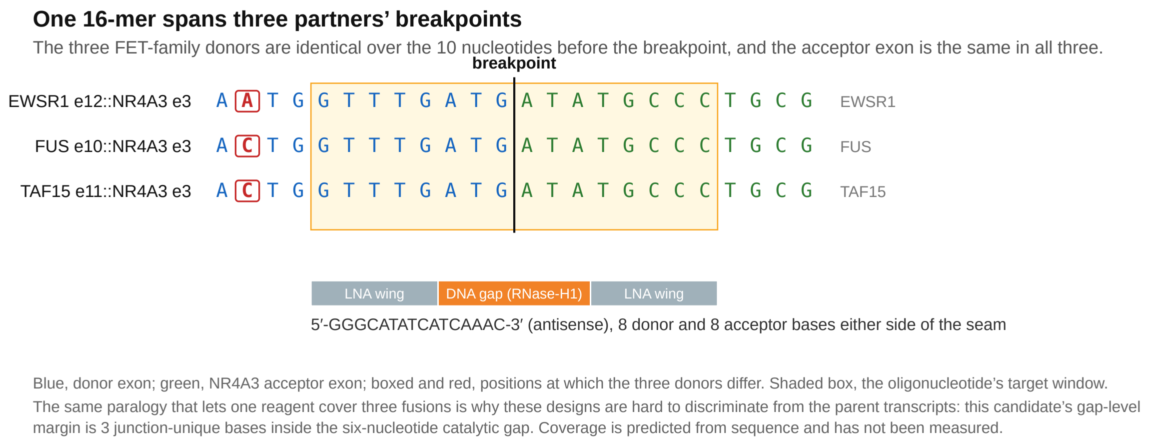
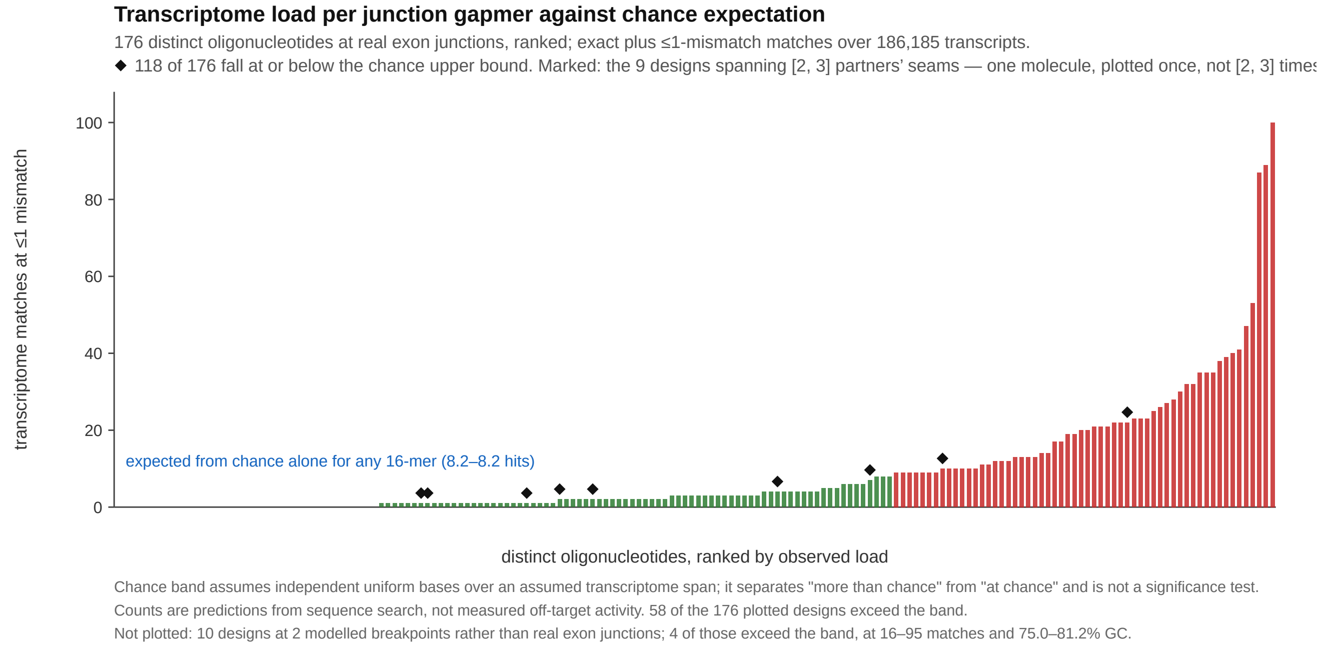


Figure 3. One 16-mer spans three partners' breakpoints. The junction windows of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, aligned at the breakpoint. Blue, donor exon; green, acceptor exon; positions at which the three donors differ are boxed as well as coloured, for greyscale and colour-blind readers. The shaded box is the target window of 5′-GGGCATATCATCAAAC-3′, with the 5-6-5 gapmer architecture below it and its gap-level margin of three alongside. The three donors are identical over the ten nucleotides before the breakpoint, which is what makes one oligonucleotide junction-spanning at all three junctions. Coverage is predicted from sequence and has not been measured.



Supplementary Figure S1. Transcriptome load per design against chance expectation. Each bar is one distinct oligonucleotide's count of exact plus ≤1-mismatch matches over 186,185 transcripts, ranked. The 190 design records at real exon junctions collapse to 176 molecules, because nine of the 16-mers are junction-spanning at more than one partner's junction at once — five at three junctions and four at two — and each of those is one physical oligonucleotide, plotted once rather than repeatedly (marked). The line is the number of such matches expected for an arbitrary 16-mer under an independent-uniform-base null, 8.2, computed against the scan's measured 718,571,139-nucleotide span; 118 of the 176 fall at or below it and 58 exceed it. Ten further designs from modelled breakpoints not built from a spliced transcript model are excluded, and are released with the artefacts. It is an expected count: the observed mean is 9.2, a ratio of 1.12, while the median is 3, so real transcript sequence produces a long right tail the null cannot rather than a uniform shift away from it. The line separates "more than chance" from "at chance" and is not a significance test; the counts are predictions from sequence search, not measured off-target activity.

Declarations

Data and code availability. [ARCHIVE DOI], deposited from `github.com/trimcrae/Rare-cancers`. A manifest listing every archived file with its SHA-256 travels with the deposit. Artefacts include the graded junction atlas, per-junction design panels, all five screens, the per-junction reagent table behind Table 4, the graded re-scores under both discrimination bounds, and the retrieval records for every literature claim.

Provenance and corrections. An earlier version of these analyses placed the acceptor junction incorrectly through a coding-versus-transcript exon indexing error and was withdrawn in full; all panels were rebuilt and verified against two independent transcript acquisitions. The complete correction record, including every superseded value, is released with the archive.

Because that is the failure a reader will reasonably assume could recur, the two instruments the paper's conclusions rest on were reimplemented a second time and the two implementations compared. The reimplementation shares no code with the original and differs from it on four axes: it splices each mature transcript out of the genomic record rather than reading the cDNA record; it locates each gene's coding start by open-reading-frame search rather than by reading the annotated 5′ untranslated length, which is the class of value the retracted error turned on; it grades the reading frame by arithmetic on exon coding-length vectors rather than by translating the chimera; and it computes the mature-parent screen by substring search over the design's gap-containing substrings rather than by scanning every parent window and extending outward from the gap. The two agree on all 231 graded exon pairs, field by field and not only on the grade, and on the longest parent duplex of all 190 designs, giving the same 87 and the same 61 against *NR4A3*. The two transcript acquisitions agree base for base for all six genes, and the annotation-free coding start reproduces the annotated one for all six. Both implementations, the comparison and its deliberate-corruption tests are in the archive.

Two things that check should not be read as. It is not external review: the same author prepared both implementations. And it bounds implementation error only — two implementations of a specification that is itself wrong will agree with each other and both be wrong, so agreement here is not evidence that the longest contiguous run containing the catalytic gap is the right quantity to compute. Independent review of the code by another group remains wanted and is not claimed.

Competing interests. The author declares no financial competing interests: he holds no position, equity, consultancy or patent relating to any gene, sequence or agent named here, and no oligonucleotide described in this manuscript has been synthesised, licensed or offered for sale. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses.

Funding. This work received no external funding and was self-funded by the author. No grant, institution, company or charity supported it, and no funder had any role in the design of the analyses, the interpretation of the results, or the decision to publish.

Ethics. No human subjects, human material or animals were involved. All clinical figures are taken from published reports and are cited.

Use of AI tools. A large language model (Claude, Anthropic) was used throughout this work: to write the analysis code, to run the graded design and screening pipelines, to draft and revise this manuscript, and to conduct internal critical review of earlier drafts. Every quantitative statement derived from sequence or from a screen is produced by code in the released archive and is reproducible from it, while the clinical figures are transcribed from the publications cited for them; no numerical result was generated by a language model directly. Every literature identifier was checked against a retrieved bibliographic record, and identifiers that could not be so anchored were removed rather than retained. The three references dated 2026 were additionally checked for live resolution against two independent registries, Europe PMC and Crossref, which return the same title and the same digital object identifier for each; those records travel with the archive. The frame grading and the mature-parent screen were additionally reimplemented and cross-checked as described under Provenance, which bounds implementation error but is not independent review. The author takes full responsibility for the content, including for the correctness of the code and for the interpretation of the results.

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52 entries, numbered by first citation in the submission manuscript. Metadata is read from retrieved bibliographic records; an unretrieved field is left absent rather than completed.

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